

2-9-10

1. Let h be a function defined for all $x \neq 0$ such that $h(4) = -3$ and the derivative of h is given by $h'(x) = \frac{x^2 - 2}{x}$ for all $x \neq 0$.

On what intervals, if any, is the graph of h concave up? Justify your answer.

Part B

1 point is earned for: $h''(x)$

1 point is earned for: $h''(x) > 0$

1 point is earned for the answer

$h''(x) = 1 + \frac{2}{x^2} > 0$ for all $x \neq 0$. Therefore the graph of h is concave up for all $x \neq 0$.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for: $h''(x)$

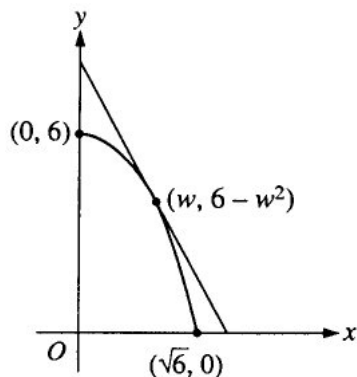
1 point is earned for: $h''(x) > 0$

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$h''(x) = 1 + \frac{2}{x^2} > 0$ for all $x \neq 0$. Therefore the graph of h is concave up for all $x \neq 0$.

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2.



Let $f(x) = 6 - x^2$. For $0 < w < \sqrt{6}$, let $A(w)$ be the area of the triangle formed by the coordinate axes and the line tangent to the graph of f at the point $(w, 6 - w^2)$.

(a) Find $A(1)$.

(b) For what value of w is $A(w)$ a minimum?

Part A

1 point is earned for the correct answer $f'(1) = -2$

$$f(x) = 6 - x^2; f'(x) = -2x$$

$$f'(1) = -2$$

1 point is earned for correctly finding equation of line or x intercept or y intercept

$$y - 5 = -2(x - 1) \text{ or } y = -2x + 7$$

x -intercept: $7/2$ y -intercept: 7

1 point is earned for the correct answer

$$A(1) = \frac{1}{2} \left(\frac{7}{2} \right) (7) = \frac{49}{4}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the correct answer $f'(1) = -2$

$$f(x) = 6 - x^2; f'(x) = -2x$$

$$f'(1) = -2$$

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$$y - 5 = -2(x - 1) \text{ or } y = -2x + 7$$

x -intercept: $7/2$ y -intercept: 7

1 point is earned for the correct answer

$$A(1) = \frac{1}{2} \left(\frac{7}{2} \right) (7) = \frac{49}{4}$$

Part B

1 point is earned for the correct equation of line

$$f'(w) = -2w; \quad y - (6 - w^2) = -2w(x - w)$$

1 point is earned for correctly expressing x intercept or y intercept in terms of w

$$x\text{-intercept} : \frac{6+w^2}{2w}; \quad y\text{-intercept} : 6 + w^2$$

1 point is earned for the correct answer for $A(w)$

$$A(w) = \frac{(6+w^2)^2}{4w}$$

1 point is earned for the correct answer for $A'(w)$

$$A'(w) = \frac{4w(2(6+w^2)(2w)) - 4(6+w^2)^2}{16w^2}$$

1 point is earned for correctly solving $A'(w) = 0$

$$A'(w) = 0 \text{ When } (6 + w^2)(3w^2 - 6) = 0$$

$$w = \sqrt{2}$$

1 point is earned for correctly showing solution yields a minimum



Note: $A(w)$ must be of the form $\frac{P(w)}{Q(w)}$, neither constant, to be eligible for derivative point

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OR

If student is using Theorem that $A(w)$ is minimized at w_1 , where $f(w_1)$ is the average value of the function:

1 point is earned for the correct average value set-up

1 point is earned for the correct antidifferentiation and evaluation

1 point is earned for correctly setting $6 - w^2 = \text{average value}$

1 point is earned for correctly solving for w_1 in domain $(0, \sqrt{6})$

2 points are earned for the clear statement of Theorem

<-1> error in statement

<-2> No statement



0	1	2	3	4	5	6
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The student response earns all of the following points:

1 point is earned for the correct equation of line

$$f'(w) = -2w; \quad y - (6 - w^2) = -2w(x - w)$$

1 point is earned for correctly expressing x intercept or y intercept in terms of w

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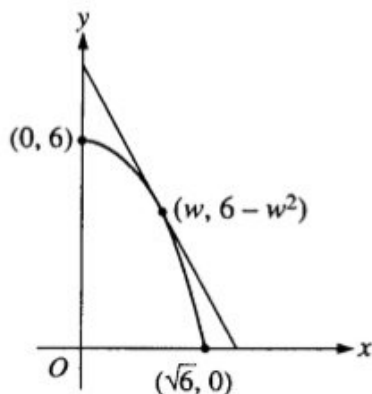
2 points are earned for the clear statement of Theorem

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2-9-10

3.



Let $f(x) = 6 - x^2$. For $0 < w < \sqrt{6}$, let $A(w)$ be the area of the triangle formed by the coordinate axes and the line tangent to the graph of f at the point $(w, 6 - w^2)$.

For what value of w is $A(w)$ a minimum?

Part B

1 point is earned for the correct equation of line

$$f'(w) = -2w; y - (6 - w^2) = -2w(x - w)$$

1 point is earned for correctly expressing x intercept or y intercept in terms of w

$$x - \text{intercept} : \frac{6 + w^2}{2w}; y - \text{intercept} : 6 + w^2$$

1 point is earned for the correct answer for $A(w)$

$$A(w) = \frac{(6 + w^2)^2}{4w}$$

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1 point is earned for correctly solving $A'(w) = 0$

$$A'(w) = 0 \quad (6 + w^2) \quad \text{when} \quad (3w^2 - 6) = 0$$

$$w = \sqrt{2}$$

1 point is earned for correctly showing solution yields a minimum



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Note: $A(w)$ must be of the form $\frac{P(w)}{Q(w)}$, neither constant, to be eligible for derivative point

OR

If student is using Theorem that $A(w)$ is minimized at w_1 , where $f(w_1)$ is the average value of the function:

1 point is earned for the correct average value set-up

1 point is earned for the correct antidifferentiation and evaluation

1 point is earned for correctly setting $6 - w^2 = \text{average value}$

1 point is earned for correctly solving for w_1 in domain $(0, \sqrt{6})$

2 points are earned for the clear statement of Theorem

< -1 > error in statement

< -2 > no statement



0	1	2	3	4	5	6
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The student response earns six of the following points:

1 point is earned for the correct equation of line

$$f'(w) = -2w; y - (6 - w^2) = -2w(x - w)$$

1 point is earned for correctly expressing x intercept or y intercept in terms of w

$$x - \text{intercept} : \frac{6+w^2}{2w}; y - \text{intercept} : 6 + w^2$$

1 point is earned for the correct answer for $A(w)$

$$A(w) = \frac{(6+w^2)^2}{4w}$$

1 point is earned for the correct answer for $A'(w)$

$$A'(w) = \frac{4w(2(6+w^2)(2w)) - 4(6+w^2)^2}{16w^2}$$

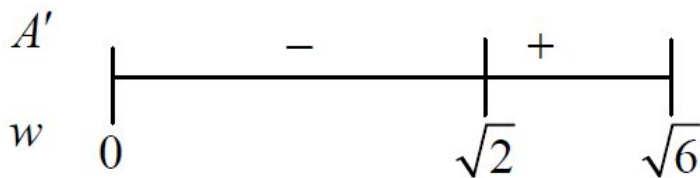
1 point is earned for correctly solving $A'(w) = 0$

$$A'(w) = 0 \quad (6 + w^2) \quad \text{when} \quad (3w^2 - 6) = 0$$

$$w = \sqrt{2}$$

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1 point is earned for correctly showing solution yields a minimum



Note: $A(w)$ must be of the form $\frac{P(w)}{Q(w)}$, neither constant, to be eligible for derivative point

OR

If student is using Theorem that $A(w)$ is minimized at w_1 , where $f(w_1)$ is the average value of the function:

1 point is earned for the correct average value set-up

1 point is earned for the correct antidifferentiation and evaluation

1 point is earned for correctly setting $6 - w^2 = \text{average value}$

1 point is earned for correctly solving for w_1 in domain $(0, \sqrt{6})$

2 points are earned for the clear statement of Theorem

< -1 > error in statement

< -2 > no statement

4. An equation of the line tangent to the graph of $y = \frac{2x+3}{3x-2}$ at the point (1, 5) is

(A) $13x - y = 8$

(B) $13x + y = 18$

(C) $x - 13y = 64$

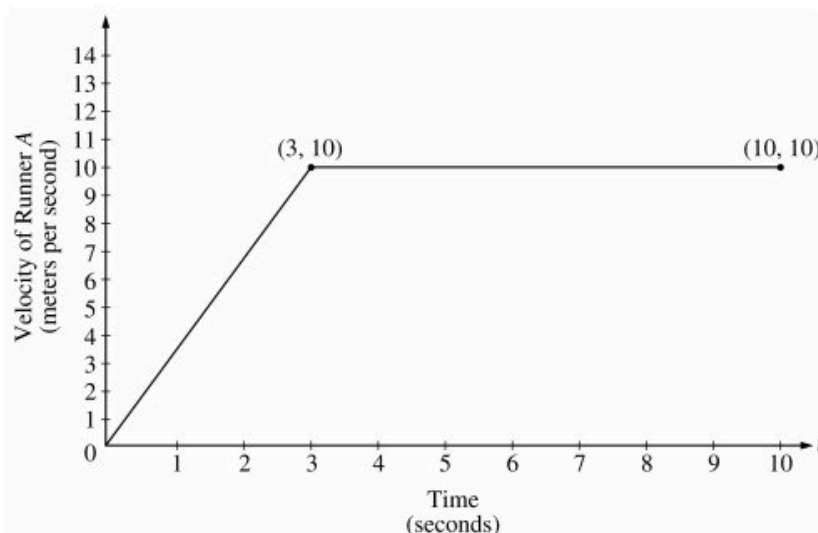
(D) $x + 13y = 66$

(E) $-2x + 3y = 13$



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5.



Two runners, A and B , run on a straight racetrack for $0 \leq t \leq 10$ seconds. The graph above, which consists of two line segments, shows the velocity, in meters per second, of Runner A . The velocity, in meters per second, of Runner B is given by the function v defined by $v(t) = \frac{24t}{2t+3}$.

Find the acceleration of Runner A and the acceleration of Runner B at time $t = 2$ seconds. Indicate the units of measure.

Part B

1 point is earned for the acceleration for Runner A

1 point is earned for the acceleration for Runner B

Runner A : $acceleration = \frac{10}{3} = 3.333meters/sec^2$

Runner B : $a(2) = v'(2) = \frac{72}{(2t+3)^2} \Big|_{t=2}$
 $= \frac{72}{49} = 1.469meters/sec^2$



0	1	2
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The student response earns all of the following points:

1 point is earned for the acceleration for Runner A

1 point is earned for the acceleration for Runner B

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Runner *A*: $acceleration = \frac{10}{3} = 3.333meters/sec^2$

Runner *B*: $a(2) = v'(2) = \frac{72}{(2t+3)^2} \Big|_{t=2}$
 $= \frac{72}{49} = 1.469meters/sec^2$

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6. Let f be the function that is given by $f(x) = ax + b/x^2 - c$ and that has the following properties.

(i) The graph of f is symmetric with respect to the y -axis.

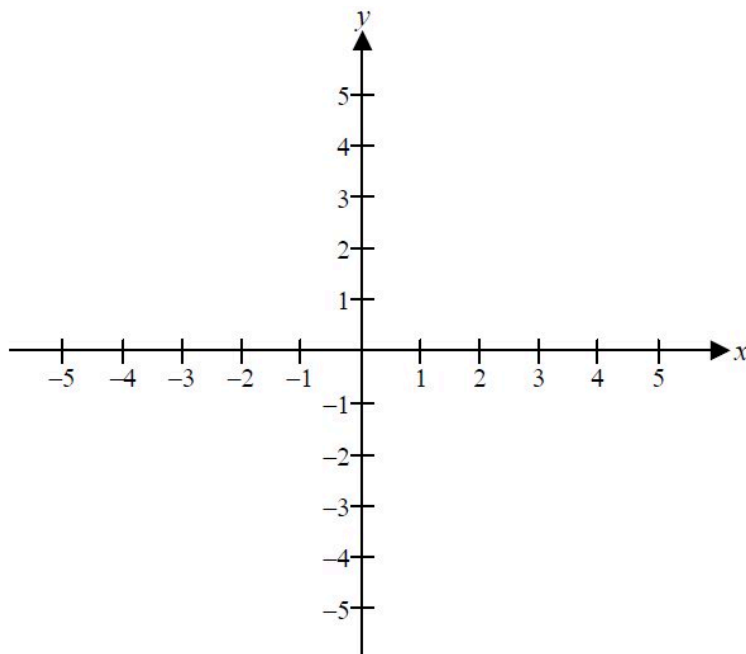
(ii) $\lim_{x \rightarrow 2^+} f(x) = +\infty$

(iii) $f'(1) = -2$

(a) Determine the values of a , b , and c .

(b) Write an equation for each vertical and each horizontal asymptote of the graph of f .

(c) Sketch the graph of f in the xy -plane provided below.

**Part A**

1 point is earned for $a = 0$

1 point is earned for $c = 4$

1 point is earned for $f'(x)$

1 point is earned for finding $b = 5$

If graph symmetric to y -axis then f is even

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$$f(-x) = f(x) \quad \therefore \boxed{a = 0}$$

$$\lim_{x \rightarrow 2^+} f(x) = +\infty \quad \therefore \boxed{c = 4}$$

$$f(x) = \frac{b}{x^2 - 4}$$

$$f'(x) = \frac{-2bx}{(x^2 - 4)^2}$$

$$-2 = f'(1) = \frac{-2b}{9} \quad \therefore \boxed{b = 9}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $a = 0$

1 point is earned for $c = 4$

1 point is earned for $f(x)$

1 point is earned for finding $b = 5$

If graph symmetric to y -axis then f is even

$$f(-x) = f(x) \quad \therefore \boxed{a = 0}$$

$$\lim_{x \rightarrow 2^+} f(x) = +\infty \quad \therefore \boxed{c = 4}$$

$$f(x) = \frac{b}{x^2 - 4}$$

$$f'(x) = \frac{-2bx}{(x^2 - 4)^2}$$

$$-2 = f'(1) = \frac{-2b}{9} \quad \therefore \boxed{b = 9}$$

Part B

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if asymptotes are switched

$$f(x) = \frac{9}{x^2 - 4}$$

2-9-10VERTICAL: $x = 2$, $x = -2$ HORIZONTAL: $y = 0$

✓

0	1	2
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The student response earns all of the following points:

1 point is earned for two vertical asymptote

1 point is earned for horizontal asymptote

Note: 1/2 points given if asymptotes are switched

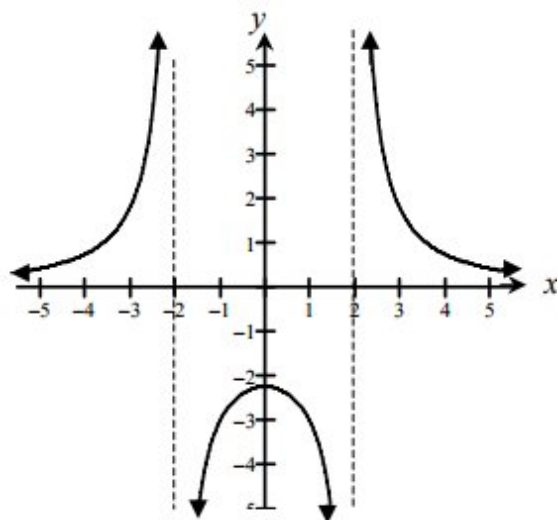
$$f(x) = \frac{9}{x^2 - 4}$$

VERTICAL: $x = 2$, $x = -2$ HORIZONTAL: $y = 0$ **Part C**

1 point is earned if symmetry agrees with their (a)

1 point is earned if asymptotes agrees with (b), (ii) verticals 2 sided

1 point is earned if shape agrees with (a), (iii), y-int



✓

0	1	2	3
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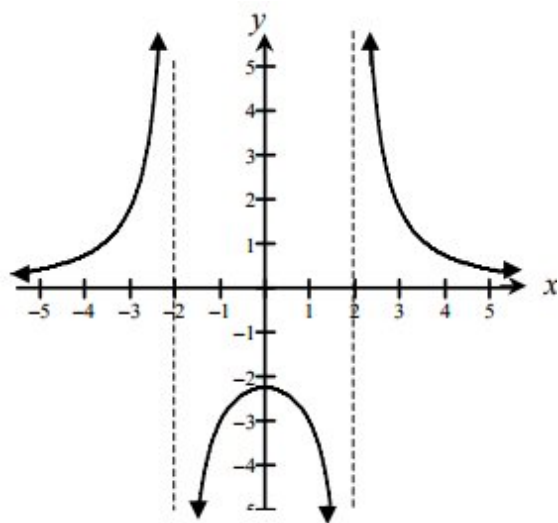
2-9-10

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1 point is earned if symmetry agrees with their (a)

1 point is earned if asymptotes agrees with (b), (ii) verticals 2 sided

1 point is earned if shape agrees with (a), (iii), y -int



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7. Let f be the function that is given by $f(x) = ax + b/x^2 - c$ and that has the following properties.

(i) The graph of f is symmetric with respect to the y -axis.

(ii) $\lim_{x \rightarrow 2^+} f(x) = +\infty$

(iii) $f'(1) = -2$

Determine the values of a , b , and c .

Part A

1 point is earned for $a=0$

1 point is earned for $c=4$

1 point is earned for $f(x)$

1 point is earned for finding $b=5$

If graph symmetric to y -axis then f is even

$f(-x) = f(x)$ therefore $a = 0$

$\lim_{x \rightarrow 2^+} f(x) = +\infty$ therefore $c = 4$

$$f(x) = \frac{b}{x^2 - 4}$$

$$f'(x) = \frac{-2bx}{(x^2 - 4)^2}$$

$$-2 = f'(1) = \frac{-2b}{9} \text{ therefore } b = 9$$



0	1	2	3	4
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The student response earns four of the following points:

1 point is earned for $a=0$

1 point is earned for $c=4$

1 point is earned for $f(x)$

1 point is earned for finding $b=5$

If graph symmetric to y -axis then f is even

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$$f(-x) = f(x) \text{ therefore } a = 0$$

$$\lim_{x \rightarrow 2^+} f(x) = +\infty \text{ therefore } c = 4$$

$$f(x) = \frac{b}{x^2 - 4}$$

$$f'(x) = \frac{-2bx}{(x^2 - 4)^2}$$

$$-2 = f'(1) = \frac{-2b}{9} \text{ therefore } b = 9$$

8. If $f(x) = \frac{3x-2}{2x+3}$, then $f'(x) =$

(A) $-\frac{13}{(2x+3)^2}$

(B) $\frac{3}{(2x+3)^2}$

(C) $\frac{5}{(2x+3)^2}$

(D) $\frac{13}{(2x+3)^2}$ ✓

(E) $\frac{12x+5}{(2x+3)^2}$

9. If $y = \frac{2x+3}{3x+2}$, then $\frac{dy}{dx} =$

(A) $\frac{12x+13}{(3x+2)^2}$

(B) $\frac{12x-13}{(3x+2)^2}$

(C) $\frac{5}{(3x+2)^2}$

(D) $\frac{-5}{(3x+2)^2}$ ✓

(E) $\frac{2}{3}$

10. What is the instantaneous rate of change at $x=2$ of the function f given by $f(x) = \frac{x^2-2}{x-1}$?

(A) -2

(B) $\frac{1}{6}$

(C) $\frac{1}{2}$

(D) 2 ✓

(E) 6

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11. If $y = \frac{\ln x}{x}$, then $\frac{dy}{dx} =$

(A) $\frac{1}{x}$

(B) $\frac{1}{x^2}$

(C) $\frac{\ln x - 1}{x^2}$

(D) $\frac{1 - \ln x}{x^2}$ ✓

(E) $\frac{1 + \ln x}{x^2}$

12. If $f(x) = \frac{5-x}{x^3+2}$, then $f'(x) =$

(A) $\frac{-4x^3+15x^2-2}{(x^3+2)^2}$

(B) $\frac{-2x^3+15x^2+2}{(x^3+2)^2}$

(C) $\frac{2x^3-15x^2-2}{(x^3+2)^2}$ ✓

(D) $\frac{4x^3-15x^2+2}{(x^3+2)^2}$

13. If $y = \tan x - \cot x$, then $dy/dx =$

(A) $\sec x \csc x$

(B) $\sec x - \csc x$

(C) $\sec x + \csc x$

(D) $\sec^2 x - \csc^2 x$

(E) $\sec^2 x + \csc^2 x$ ✓

Answer E

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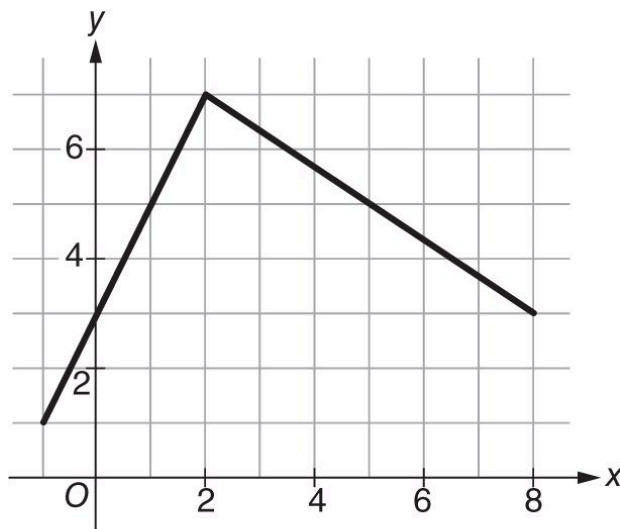
14.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Graph of f

Let f be the continuous function defined on $[-1, 8]$ whose graph, consisting of two line segments, is shown above. Let g and h be the functions defined by $g(x) = \sqrt{x^2 - x + 3}$ and $h(x) = 5e^x - 9 \sin x$.

- (a) The function k is defined by $k(x) = f(x)g(x)$. Find $k'(0)$.
- (b) The function m is defined by $m(x) = \frac{f(x)}{2g(x)}$. Find $m'(5)$.
- (c) Find the value of x for $-1 < x < 2$ such that $f'(x) = h'(x)$.

Part A

The third point is earned for a correct product rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified. The third point can also be earned if a student imports an incorrect value for either $f'(0)$ or $g'(0)$.

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

☐ $f'(0)$

☐ $g'(0)$

☐ $k'(0)$

Solution:

$$k'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

$$f'(0) = 2 \text{ from the graph.}$$

$$g'(0) = -0.288675$$

$$k'(0) = f'(0) \cdot g(0) + f(0) \cdot g'(0)$$

$$= 2 \cdot \sqrt{3} + 3 \cdot (-0.288675) = 2.598$$

Part B

The third point is earned for a correct quotient rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

☐ $f'(5)$

☐ $g'(5)$

☐ $m'(5)$

Solution:

$$m'(x) = \frac{f'(x) \cdot 2g(x) - f(x) \cdot 2g'(x)}{(2g(x))^2}$$

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$f'(5) = -\frac{2}{3}$ from the graph.

$$g'(5) = 0.938315$$

$$m'(5) = \frac{f'(5) \cdot 2g(5) - f(5) \cdot 2g'(5)}{(2g(5))^2}$$

$$= \frac{\left(-\frac{2}{3}\right) \cdot 2\sqrt{23} - 5 \cdot 2 \cdot 0.938315}{\left(2\sqrt{23}\right)^2} = -0.171$$

Part C

If $h'(x)$ has a maximum of one sign error with either term, the response is eligible for the second point based on a correct evaluation with the presented $h'(x)$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

☐ $h'(x)$

☐ answer

Solution:

$$f'(x) = 2 \text{ for } -1 < x < 2$$

$$h'(x) = 5e^x - 9 \cos x$$

$$h'(x) = 2 \Rightarrow x = 0.622$$

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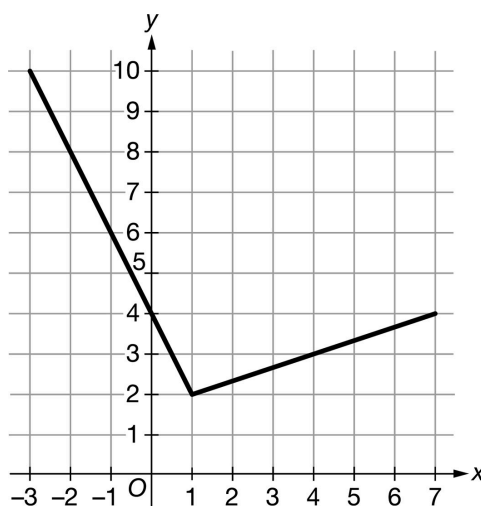
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Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

Let f be the continuous function defined on $[-3, 7]$ whose graph, consisting of two line segments, is shown above. Let g and h be the functions defined by $g(x) = (x^2 + 5x)^{\frac{1}{3}}$ and $h(x) = 7\cos x + x^3$.

- The function N is defined by $N(x) = f(x)g(x)$. Find $N'(-1)$. Show the work that leads to your answer.
- The function P is defined by $P(x) = \frac{g(x)}{5f(x)}$. Find $P'(4)$. Show the work that leads to your answer.
- Find the value of x for $-3 < x < 1$ such that $f'(x) = h'(x)$.

Part A

The third point is earned for a correct product rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified. The third point can also be earned if a student imports an incorrect value for either $f'(-1)$ or $g'(-1)$.

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $f'(-1)$
- ☐ $g'(-1)$.
- ☐ $N'(-1)$

Solution:

$$N'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$$

$f'(-1) = -2$ determined from the slope of the line segment on the graph.

$$g'(-1) = 0.396850$$

$$\begin{aligned} N'(-1) &= f'(-1) \cdot g(-1) + f(-1) \cdot g'(-1) \\ &= (-2) \cdot (-4)^{\frac{1}{3}} + 6 \cdot 0.396850 = 5.556 \text{ (or } 5.555) \end{aligned}$$

Part B

The third point is earned for a correct quotient rule expression that substitutes all 4 correct numerical values. The expression does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ $f'(4)$
- ☐ $g'(4)$
- ☐ $P'(4)$

Solution:

$$P'(x) = \frac{g'(x) \cdot 5f(x) - g(x) \cdot 5f'(x)}{(5f(x))^2}$$

2-9-10

$f'(4) = \frac{1}{3}$ determined from the slope of the line segment on the graph.

$$g'(4) = 0.397454$$

$$P'(4) = \frac{g'(4) \cdot 5f(4) - g(4) \cdot 5f'(4)}{(5f(4))^2}$$

$$= \frac{0.397454 \cdot 5 \cdot 3 - 36^{\frac{1}{3}} \cdot 5 \cdot \frac{1}{3}}{(5 \cdot 3)^2} = 0.002$$

Part C

If $h'(x)$ has a maximum of one sign error with either term, the response is eligible for the second point based on a correct evaluation with the presented $h'(x)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $h'(x)$

☐ answer

Solution:

$f'(x) = -2$ determined from the slope of the line segment on the graph for $-3 < x < 1$

$$h'(x) = -7 \sin x + 3x^2$$

$$h'(x) = -2 \Rightarrow x = 0.343 \text{ (or } 0.342)$$

2-9-10

16. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	-2	-1	1
$f(x)$	$-\frac{5}{2}$	-3	-2	$\frac{2}{3}$
$f'(x)$	-1	$\frac{1}{3}$	$\frac{6}{5}$	$\frac{4}{3}$

The table above gives values of the differentiable function f and its derivative for selected values of x .

- (a) Let g be the function defined by $g(x) = \frac{f(x^2)}{e^x}$. Find $g'(-1)$.
- (b) Let h be the function defined by $h(x) = f(f(-2x))$. Find $h'(1)$.
- (c) Let k be the function defined by $k(x) = f(x) \cdot \arcsin\left(\frac{x}{2}\right)$. Find $k'(-1)$.

Part A

The first and second points require evidence of quotient rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of quotient rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student responses accurately includes all three of the criteria below.

- ☐ quotient rule
- ☐ chain rule
- ☐ answer

Solution:

2-9-10

$$g'(x) = \frac{f'(x^2) \cdot 2x \cdot e^x - f(x^2) \cdot e^x}{(e^x)^2}$$

$$\begin{aligned} g'(-1) &= \frac{f'(1) \cdot 2(-1) \cdot e^{-1} - f(1) \cdot e^{-1}}{(e^{-1})^2} \\ &= \frac{\frac{4}{3}(-2)e^{-1} - \frac{2}{3}e^{-1}}{(e^{-1})^2} = -\frac{10e}{3} \end{aligned}$$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $h'(x)$
- ☐ answer

Solution:

$$\begin{aligned} h'(x) &= f'(f(-2x)) \cdot f'(-2x) \cdot (-2) \\ h'(1) &= f'(f(-2)) \cdot f'(-2) \cdot (-2) \\ &= f'(-3) \cdot \frac{1}{3} \cdot (-2) = (-1) \cdot \frac{1}{3} \cdot (-2) = \frac{2}{3} \end{aligned}$$

Part C

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required; trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

2-9-10

The student response accurately includes all three of the criteria below.

- ☐ product rule
- ☐ chain rule
- ☐ answer

Solution:

$$k'(x) = f'(x) \cdot \arcsin\left(\frac{x}{2}\right) + f(x) \cdot \frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}} \cdot \frac{1}{2}$$

$$\begin{aligned} k'(-1) &= f'(-1) \cdot \arcsin\left(\frac{-1}{2}\right) + f(-1) \cdot \frac{1}{\sqrt{1-\left(\frac{-1}{2}\right)^2}} \cdot \frac{1}{2} \\ &= \frac{6}{5} \cdot \left(-\frac{\pi}{6}\right) + (-2) \cdot \frac{2}{\sqrt{3}} \cdot \frac{1}{2} = -\frac{\pi}{5} - \frac{2}{\sqrt{3}} \end{aligned}$$

2-9-10

17.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	$f(x)$	$f'(x)$
3	-1	8
5	4	3
7	6	-2

The table above gives values of the differentiable function f and its derivative f' at selected values of x . Let h be the function defined by $h(x) = x^2 e^{\cos(3x)}$.

- Using values of f from the table, estimate $f'(4)$. Show the work that leads to your answer.
- Let M be the function defined by $M(x) = f(x) + h(x)$. Find $M'(3)$.
- Let N be the function defined by $N(x) = f(x)h(x)$. Find $N'(5)$.
- Let P be the function defined by $P(x) = \frac{h(x)}{f(x)}$. Find $P'(7)$.

Part A

The point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified. If simplified, there are no errors. A response that presents an incorrect expression, makes an error in substituting values from the table, or makes an error in simplification does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

2-9-10

The student response accurately includes a correct estimate of $f'(4)$

Solution:

$$f'(4) \approx \frac{f(5)-f(3)}{5-3} = \frac{4-(-1)}{5-3} = \frac{5}{2}$$

Part B

The first point is earned for correctly presenting the rule for derivative of a sum, either by writing $M'(x) = f'(x) + h'(x)$ or $M'(3) = f'(3) + h'(3)$ or $M'(3) = 8 + (-2.061490)$ OR for the correct answer accompanied by $f'(3) = 8$ and $h'(3) = -2.061490$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $8 + (-2.061490)$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $M'(x)$

☐ answer

Solution:

$$M'(x) = f'(x) + h'(x)$$

$$f'(3) = 8 \text{ from the table.}$$

$$h'(3) = -2.061490 \text{ from the calculator.}$$

$$M'(3) = f'(3) + h'(3) = 5.939 \text{ (or } 5.938)$$

Part C

The first point is earned for correctly presenting the product rule for derivatives, either by writing $N'(x) = f'(x)h(x) + f(x)h'(x)$ or $N'(5) = f'(5)h(5) + f(5)h'(5)$ or $N'(5) = 3(11.695310) + 4(-18.137830)$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $3(11.695310) + 4(-18.137830)$ earns the second point.

2-9-10

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $N'(x)$

☐ answer

Solution:

$$N'(x) = f'(x)h(x) + f(x)h'(x)$$

$f(5) = 4$ and $f'(5) = 3$ from the table.

$h(5) = 11.695310$ and $h'(5) = -18.137830$ from the calculator.

$$N'(5) = f'(5)h(5) + f(5)h'(5) = -37.465$$

Part D

The first point is earned for correctly presenting the quotient rule for derivatives, either by writing

$$P'(x) = \frac{h'(x)f(x) - h(x)f'(x)}{(f(x))^2} \text{ or } P'(7) = \frac{h'(7)f(7) - h(7)f'(7)}{(f(7))^2} \text{ or}$$

$$P'(7) = \frac{(-63.023773) \cdot 6 - 28.334809 \cdot (-2)}{6^2}.$$

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term in the numerator and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $\frac{(-63.023773) \cdot 6 - 28.334809 \cdot (-2)}{6^2}$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $P'(x)$

☐ answer

2-9-10**Solution:**

$$P'(x) = \frac{h'(x)f(x) - h(x)f'(x)}{(f(x))^2}$$

$f(7) = 6$ and $f'(7) = -2$ from the table.

$h(7) = 28.334809$ and $h'(7) = -63.023773$ from the calculator.

$$P'(7) = \frac{h'(7)f(7) - h(7)f'(7)}{(f(7))^2} = -8.930 \text{ (or } -8.929)$$

2-9-10

18.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	$g(x)$	$g'(x)$
2	-2	4
4	5	1
6	2	-3

The table above gives values of the differentiable function g and its derivative g' at selected values of x . Let f be the function defined by $f(x) = x \sin(\sqrt{3x^2 + 5})$.

- Using values of g from the table, estimate $g'(5)$. Show the work that leads to your answer.
- Let R be the function defined by $R(x) = f(x) - g(x)$. Find $R'(2)$.
- Let S be the function defined by $S(x) = f(x)g(x)$. Find $S'(4)$.
- Let T be the function defined by $T(x) = \frac{f(x)}{g(x)}$. Find $T'(6)$.

Part A

The point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified. If simplified, there are no errors. A response that presents an incorrect expression, makes an error in substituting values from the table, or makes an error in simplification does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

2-9-10

The student response accurately includes a correct estimate of $g'(5)$

Solution:

$$g'(5) \approx \frac{g(6)-g(4)}{6-4} = \frac{2-5}{6-4} = -\frac{3}{2}$$

Part B

The first point is earned for correctly presenting the rule for derivative of a difference, either by writing $R'(x) = f'(x) - g'(x)$ or $R'(2) = f'(2) - g'(2)$ or $R'(2) = -2.448854 - 4$ OR for the correct answer accompanied by $f'(2) = -2.448854$ and $g'(2) = 4$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $-2.448854 - 4$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $R'(x)$

☐ answer

Solution:

$$R'(x) = f'(x) - g'(x)$$

$$f'(2) = -2.448854 \text{ from the calculator.}$$

$$g'(2) = 4 \text{ from the table.}$$

$$R'(2) = f'(2) - g'(2) = -6.449$$

Part C

The first point is earned for correctly presenting the product rule for derivatives, either by writing $S'(x) = f'(x)g(x) + f(x)g'(x)$ or $S'(4) = f'(4)g(4) + f(4)g'(4)$ or

$$S'(4) = (4.419230) \cdot (5) + (3.359221) \cdot (1).$$

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn

2-9-10

any points. An unsimplified final answer such as

$(4.419230) \cdot (5) + (3.359221) \cdot (1)$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $S'(x)$

☐ answer

Solution:

$$S'(x) = f'(x)g(x) + f(x)g'(x)$$

$f(4) = 3.359221$ and $f'(4) = 4.419230$ from the calculator.

$g(4) = 5$ and $g'(4) = 1$ from the table.

$$S'(4) = f'(4)g(4) + f(4)g'(4) = 25.455$$

Part D

The first point is earned for correctly presenting the quotient rule for derivatives, either by writing

$$T'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \text{ or } T'(6) = \frac{f'(6)g(6) - f(6)g'(6)}{(g(6))^2} \text{ or}$$

$$T'(6) = \frac{(-4.564565) \cdot (2) - (-5.603824) \cdot (-3)}{(2)^2}.$$

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term in the numerator and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as

$$\frac{(-4.564565) \cdot (2) - (-5.603824) \cdot (-3)}{(2)^2} \text{ earns the second point.}$$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

2-9-10

The student response accurately includes both of the criteria below.

☐ $T'(x)$

☐ answer

Solution:

$$T'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

$$f(6) = -5.603824 \text{ and}$$

$$f'(6) = -4.564565 \text{ from the calculator.}$$

$$g(6) = 2 \text{ and } g'(6) = -3 \text{ from the table.}$$

$$T'(6) = \frac{f'(6)g(6) - f(6)g'(6)}{(g(6))^2} = -6.485$$

2-9-10

19. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-3	2	3	8
$f(x)$	-9	4	2	6
$f'(x)$	$-\frac{7}{2}$	$\frac{3}{2}$	$-\frac{2}{5}$	$\frac{1}{3}$

The table above gives values of the differentiable function f and its derivative for selected values of x .

- (a) Let g be the function defined by $g(x) = \frac{\ln x}{f(x^3)}$. Find $g'(2)$.
- (b) Let h be the function defined by $h(x) = f(f(-3x))$. Find $h'(-1)$.
- (c) Let k be the function defined by $k(x) = f(x) \cdot \arctan\left(\frac{x}{3}\right)$. Find $k'(-3)$.

Part A

The first and second points require evidence of quotient rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of quotient rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student responses accurately includes all three of the criteria below.

- ☐ quotient rule
- ☐ chain rule
- ☐ answer

Solution:

2-9-10

$$g'(x) = \frac{f(x^3) \cdot \frac{1}{x} - f'(x^3) \cdot 3x^2 \cdot \ln x}{(f(x^3))^2}$$

$$g'(2) = \frac{f(8) \cdot \frac{1}{2} - f'(8) \cdot 12 \ln 2}{(f(8))^2}$$

$$= \frac{6 \cdot \frac{1}{2} - \frac{1}{3} \cdot 12 \ln 2}{(6)^2} = \frac{3 - 4 \ln 2}{36}$$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response does not accurately include any of the criteria below.

- ☐ $h'(x)$
- ☐ answer

Solution:

$$h'(x) = f'(f(-3x)) \cdot f'(-3x) \cdot (-3)$$

$$h'(-1) = f'(f(3)) \cdot f'(3) \cdot (-3)$$

$$= f'(2) \cdot \left(-\frac{2}{5}\right) \cdot (-3) = \left(\frac{3}{2}\right) \cdot \left(-\frac{2}{5}\right) \cdot (-3) = \frac{9}{5}$$

Part C

The first and second points require evidence of product rule and chain rule and no errors. At most 1 out of 3 points is earned for partial communication of product rule and chain rule with a maximum of one computational error. Substitution of function values is required. A simplified answer is not required; trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

2-9-10

The student response accurately includes all three of the criteria below.

- ☐ product rule
- ☐ chain rule
- ☐ answer

Solution:

$$k'(x) = f'(x) \cdot \arctan\left(\frac{x}{3}\right) + f(x) \cdot \frac{1}{1+\left(\frac{x}{3}\right)^2} \cdot \frac{1}{3}$$

$$\begin{aligned} k'(-3) &= f'(-3) \cdot \arctan(-1) + f(-3) \cdot \frac{1}{1+\left(\frac{-3}{3}\right)^2} \cdot \frac{1}{3} \\ &= -\frac{7}{2} \cdot \left(-\frac{\pi}{4}\right) + (-9) \cdot \frac{1}{2} \cdot \frac{1}{3} = \frac{7\pi}{8} - \frac{3}{2} \end{aligned}$$

2-9-10

20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Consider the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$.

(a) Show that $\frac{d^2y}{dx^2} = \frac{2y^3 - y^2 - 2xy}{x^2}$.

(b) Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$ with initial condition $g(4) = 2$. Does g have a relative minimum, a relative maximum, or neither at $x = 4$? Justify your answer.

(c) Let $y = h(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$ with initial condition $h(1) = 2$. Write the second-degree Taylor polynomial for h about $x = 1$.

(d) For the function h given in part (c), it is known that $|h'''(x)| \leq 60$ for all x in the interval $0.9 \leq x \leq 1.1$. Let A represent the approximation of $h(1.1)$ found by using the second-degree Taylor polynomial for h about $x = 1$ from part (c). Use the Lagrange error bound to show that A differs from $h(1.1)$ by at most 0.01.

Part A

1 point is earned for: quotient rule

1 point is earned for: $\frac{d^2y}{dx^2}$

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{x \cdot 2y \frac{dy}{dx} - y^2 \cdot 1}{x^2} \\ &= \frac{2xy \left(-1 + \frac{y^2}{x}\right) - y^2}{x^2} = \frac{2y^3 - y^2 - 2xy}{x^2}\end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: quotient rule

1 point is earned for: $\frac{d^2y}{dx^2}$

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{x \cdot 2y \frac{dy}{dx} - y^2 \cdot 1}{x^2} \\ &= \frac{2xy \left(-1 + \frac{y^2}{x}\right) - y^2}{x^2} = \frac{2y^3 - y^2 - 2xy}{x^2}\end{aligned}$$

2-9-10

Part B

1 point is earned for: considers $\frac{dy}{dx}\big|_{(x,y)=(4,2)}$

1 point is earned for: answer with justification

$$\frac{dy}{dx}\big|_{(x,y)=(4,2)} = -1 + \frac{4}{4} = 0$$

$$\frac{d^2y}{dx^2}\big|_{(x,y)=(4,2)} = \frac{2 \cdot 8 - 4 - 16}{16} = -\frac{1}{4} < 0$$

By the Second Derivative Test, g has a relative maximum at $x = 4$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: considers $\frac{dy}{dx}\big|_{(x,y)=(4,2)}$

1 point is earned for: answer with justification

$$\frac{dy}{dx}\big|_{(x,y)=(4,2)} = -1 + \frac{4}{4} = 0$$

$$\frac{d^2y}{dx^2}\big|_{(x,y)=(4,2)} = \frac{2 \cdot 8 - 4 - 16}{16} = -\frac{1}{4} < 0$$

By the Second Derivative Test, g has a relative maximum at $x = 4$.

Part C

1 point is earned for: uses $\frac{dy}{dx}\big|_{(x,y)=(1,2)}$ and $\frac{d^2y}{dx^2}\big|_{(x,y)=(1,2)}$

2 points are earned for: Taylor polynomial

$$\frac{dy}{dx}\big|_{(x,y)=(1,2)} = -1 + \frac{4}{1} = 3$$

$$\frac{d^2y}{dx^2}\big|_{(x,y)=(1,2)} = \frac{2 \cdot 8 - 4 - 4}{1} = 8$$

The second-degree Taylor polynomial for h about $x = 1$ is

$$T_2(x) = 2 + 3(x - 1) + \frac{8}{2!}(x - 1)^2 = 2 + 3(x - 1) + 4(x - 1)^2.$$

2-9-10



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: uses $\frac{dy}{dx}\big|_{(x,y)=(1,2)}$ and $\frac{d^2y}{dx^2}\big|_{(x,y)=(1,2)}$

2 points are earned for: Taylor polynomial

$$\frac{dy}{dx}\big|_{(x,y)=(1,2)} = -1 + \frac{4}{1} = 3$$

$$\frac{d^2y}{dx^2}\big|_{(x,y)=(1,2)} = \frac{2 \cdot 8 - 4 - 4}{1} = 8$$

The second-degree Taylor polynomial for h about $x = 1$ is

$$T_2(x) = 2 + 3(x - 1) + \frac{8}{2!}(x - 1)^2 = 2 + 3(x - 1) + 4(x - 1)^2.$$

Part D

1 point is earned for: form of the error bound

1 point is earned for: analysis

$$|h(1.1) - A| \leq \frac{\max_{1.0 \leq x \leq 1.1} |h'''(x)| |1.1 - 1|^3}{3!} \leq \frac{60}{6} \cdot \frac{1}{1000} = \frac{10}{1000} = \frac{1}{100}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: form of the error bound

1 point is earned for: analysis

$$|h(1.1) - A| \leq \frac{\max_{1.0 \leq x \leq 1.1} |h'''(x)| |1.1 - 1|^3}{3!} \leq \frac{60}{6} \cdot \frac{1}{1000} = \frac{10}{1000} = \frac{1}{100}$$

2-9-10

21. $\frac{d}{dx}\left(\frac{x+1}{x^2+1}\right) =$

(A) $\frac{x^2+2x-1}{(x^2+1)^2}$

(B) $\frac{-x^2-2x+1}{x^2+1}$

(C) $\frac{-x^2-2x+1}{(x^2+1)^2}$ ✓

(D) $\frac{3x^2+2x+1}{(x^2+1)^2}$

(E) $\frac{1}{2x}$

22. If $f(x) = \frac{x^2+3x+2}{x+3}$, then $f'(x) =$

(A) $2x + 3$

(B) $\frac{-x^2-6x-7}{(x+3)^2}$

(C) $\frac{x^2+6x+7}{(x+3)^2}$ ✓

(D) $\frac{x^2+12x+11}{(x+3)^2}$

(E) $\frac{3x^2+12x+11}{(x+3)^2}$

Answer C

23. The function f is defined by $f(x) = \frac{x}{x+2}$. What points (x, y) on the graph of f have the property that the line tangent to f at (x, y) has slope $\frac{1}{2}$?

(A) $(0, 0)$ only

(B) $\left(\frac{1}{2}, \frac{1}{5}\right)$ only

(C) $(0, 0)$ and $(-4, 2)$ ✓

(D) $(0, 0)$ and $\left(4, \frac{2}{3}\right)$

(E) There are no such points.

2-9-10

24. If $f(x) = \frac{5-x}{x^3+2}$, then $f'(x) =$

(A) $\frac{-4x^3+15x^2-2}{(x^3+2)^2}$

(B) $\frac{-2x^3+15x^2+2}{(x^3+2)^2}$

(C) $\frac{2x^3-15x^2-2}{(x^3+2)^2}$ ✓

(D) $\frac{4x^3-15x^2+2}{(x^3+2)^2}$

2-9-10

25. Let f be the function given by $f(x) = \frac{x}{\sqrt{x^2-4}}$.

Find $f'(x)$

Part D

1 point is earned for derivative

2 point is earned for solution

$$\begin{aligned} f'(x) &= \frac{\sqrt{x^2-4} - x \left[\frac{1}{2}(x^2-4)^{-1/2} 2x \right]}{x^2-4} \\ &= \frac{\sqrt{x^2-4} - \frac{x^2}{\sqrt{x^2-4}}}{x^2-4} \\ &= \frac{-4}{(x^2-4)^{3/2}} \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for derivative

2 point is earned for solution

$$\begin{aligned} f'(x) &= \frac{\sqrt{x^2-4} - x \left[\frac{1}{2}(x^2-4)^{-1/2} 2x \right]}{x^2-4} \\ &= \frac{\sqrt{x^2-4} - \frac{x^2}{\sqrt{x^2-4}}}{x^2-4} \\ &= \frac{-4}{(x^2-4)^{3/2}} \end{aligned}$$

26. What is the slope of the line tangent to the graph of $y = \frac{x^2-2}{x^2+1}$ when $x = 1$?

- (A) $-\frac{3}{2}$
 (B) $-\frac{1}{2}$
 (C) $\frac{1}{2}$
 (D) 1

- (E) $\frac{3}{2}$



2-9-10

27.

$f(3)$	$g(3)$	$f'(3)$	$g'(3)$
-1	2	5	-2

The table above gives values for the functions f and g and their derivatives at $x = 3$. Let k be the function given by $k(x) = \frac{f(x)}{g(x)}$, where $g(x) \neq 0$. What is the value of $k'(3)$?

(A) $-\frac{5}{2}$

(B) -2

(C) 2

(D) 3

(E) 8

28.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
0	3	4	2	π

The table above gives values of the differentiable functions f and g and their derivatives at $x = 0$. If

$h(x) = \frac{f(x)}{g(x)}$, what is the value of $h'(0)$?

(A) $\frac{8-3\pi}{4}$

(B) $\frac{3\pi-8}{4}$

(C) $\frac{4}{\pi}$

(D) $\frac{2-3\pi}{2}$

(E) $\frac{8+3\pi}{4}$

29.

Given that $3x - \tan y = 4$, what is $\frac{dy}{dx}$ in terms of y ?

(A) $\frac{dy}{dx} = 3\sin^2 y$

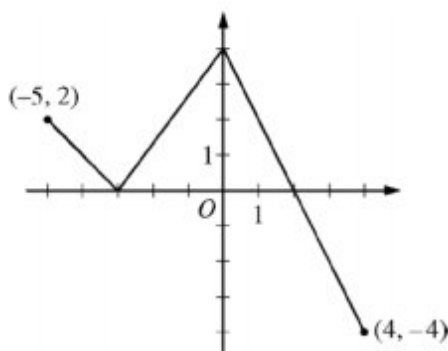
(B) $\frac{dy}{dx} = 3\cos^2 y$

(C) $\frac{dy}{dx} = 3 \cos y \cot y$

(D) $\frac{dy}{dx} = \frac{3}{1+9y^2}$

2-9-10

30.

Graph of f

The function f is defined on the closed interval $[-5, 4]$. The graph of f consists of three line segments and is shown in the figure above. Let g be the function defined by $g(x) = \int_{-3}^x f(t) dt$.

(a) Find $g(3)$.

(b) On what open intervals contained in $-5 < x < 4$ is the graph of g both increasing and concave down? Give a reason for your answer.

(c) The function h is defined by $h(x) = \frac{g(x)}{5x}$. Find $h'(3)$.

(d) The function p is defined by $p(x) = f(x^2 - x)$. Find the slope of the line tangent to the graph of p at the point where $x = -1$.

Part A

1 point is earned for the answer

$$g(3) = \int_{-3}^3 f(t) dt = 6 + 4 - 1 = 9$$



0

1

The student response earns all of the following points:

1 point is earned for the answer

$$g(3) = \int_{-3}^3 f(t) dt = 6 + 4 - 1 = 9$$

Part B

2-9-10

1 point is earned for the answer

1 point is earned for the reason

$$g'(x) = f(x)$$

The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the reason

$$g'(x) = f(x)$$

The graph of g is increasing and concave down on the intervals $-5 < x < -3$ and $0 < x < 2$ because $g' = f$ is positive and decreasing on these intervals.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = \frac{5xg'(x) - g(x)5}{(5x)^2} = \frac{5xg'(x) - 5g(x)}{25x^2}$$

$$h'(3) = \frac{(5)(3)g'(3) - 5g(3)}{25 \cdot 3^2}$$

$$= \frac{15(-2) - 5(9)}{225} = \frac{-75}{225} = -\frac{1}{3}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $h'(x)$

1 point is earned for the answer

2-9-10

$$h'(x) = \frac{5xg'(x) - g(x)5}{(5x)^2} = \frac{5xg'(x) - 5g(x)}{25x^2}$$

$$h'(3) = \frac{(5)(3)g'(3) - 5g(3)}{25 \cdot 3^2}$$

$$= \frac{15(-2) - 5(9)}{225} = \frac{-75}{225} = -\frac{1}{3}$$

Part D2 points are earned for $p'(x)$

1 point is earned for the answer

$$p'(x) = f'(x^2 - x)(2x - 1)$$

$$p'(-1) = f'(2)(-3) = (-2)(-3) = 6$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $p'(x)$

1 point is earned for the answer

$$p'(x) = f'(x^2 - x)(2x - 1)$$

$$p'(-1) = f'(2)(-3) = (-2)(-3) = 6$$

2-9-10

31. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Particle P moves along the x -axis so that its position at time $t > 0$ is given by $x_P(t) = \frac{e^{2-t}-2t}{e^{2-t}+3t}$. A second particle, particle Q , also moves along the x -axis so that its position at time t is given by $x_Q(t) = t^3 - 3t^2 + 5$.

- Show that the velocity of particle P at time t is given by $v_P(t) = \frac{-5te^{2-t}-5e^{2-t}}{(e^{2-t}+3t)^2}$.
- At time $t = 2$, particle Q is at rest. At time $t = 2$, is particle P moving toward particle Q or away from particle Q ? Explain your reasoning.
- The acceleration of particle Q is given by $a_Q(t)$. Find the value of $a_Q(2)$.
- Describe the position of particle P and the position of particle Q as t approaches infinity. Show the work that leads to your answers.

Part A

The first point is earned for successful demonstration of the quotient rule.

The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ quotient rule
- ☐ verification

Solution:

2-9-10

$$v_P(t) = x'(t) = \frac{(-e^{2-t}-2)(e^{2-t}+3t) - (e^{2-t}-2t)(-e^{2-t}+3)}{(e^{2-t}+3t)^2}$$

$$= \frac{(-e^{4-2t}-3te^{2-t}-2e^{2-t}-6t) - (-e^{4-2t}+3e^{2-t}+2te^{2-t}-6t)}{(e^{2-t}+3t)^2} = \frac{-5te^{2-t}-5e^{2-t}}{(e^{2-t}+3t)^2}$$

Part B

The response is eligible for the third point if either of the first 2 points is earned and the answer with reason is consistent with the previous results.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $x_P(2)$ and $x_Q(2)$
- ☐ $v_P(2) < 0$
- ☐ answer with explanation

Solution:

$$x_P(2) = \frac{1-4}{1+6} = -\frac{3}{7};$$

$$x_Q(2) = 1$$

$$v_P(2) = \frac{-10-5}{7^2} = -\frac{15}{49} < 0$$

Because $x_P(2) < x_Q(2)$, particle P is to the left of particle Q at time $t = 2$.

Because $v_P(2) < 0$, P is moving to the left at time $t = 2$.

Because particle Q is at rest at time $t = 2$, particle P is moving away from particle Q at time $t = 2$.

Part C

Supporting work is not required to earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-9-10



0	1
---	---

The student response accurately includes a correct value of $a_Q(2)$

$$v_Q(t) = x_Q'(t) = 3t^2 - 6t$$

$$a_Q(t) = v_Q'(t) = 6t - 6$$

$$a_Q(2) = 12 - 6 = 6$$

Part D

The first point requires showing that an indeterminate form $\frac{\infty}{\infty}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $\frac{\infty}{\infty}$ " or not using limit notation) impacts the first point.

For the second and third points, at most 1 out of 2 points may be earned for both correct limits without a connection to the interpretation of position.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ applies L'Hospital's Rule
- ☐ description of position of particle P
- ☐ description of position of particle Q

Solution:

$$\lim_{t \rightarrow \infty} e^{2-t} - 2t = -\infty$$

$$\lim_{t \rightarrow \infty} e^{2-t} + 3t = \infty$$

By L'Hospital's Rule,

$$\lim_{t \rightarrow \infty} x_P(t) = \lim_{t \rightarrow \infty} \frac{e^{2-t} - 2t}{e^{2-t} + 3t} = \lim_{t \rightarrow \infty} \frac{-e^{2-t} - 2}{-e^{2-t} + 3} = -\frac{2}{3}$$

Therefore, particle P approaches $x = -\frac{2}{3}$ as t approaches infinity.

2-9-10

$$\lim_{t \rightarrow \infty} x_Q(t) = \lim_{t \rightarrow \infty} (t^3 - 3t^2 + 5) = \infty$$

Therefore, the position of particle Q grows without bound as t approaches infinity.

2-9-10

32.  A particle moves along the x -axis with velocity given by $v(t) = \frac{10 \sin(0.4t^2)}{t^2 - t + 3}$ for time $0 \leq t \leq 3.5$. The particle is at position $x = -5$ at time $t = 0$.
- (a) Find the acceleration of the particle at time $t = 3$.
- (b) Find the position of the particle at time $t = 3$.
- (c) Evaluate $\int_0^{3.5} v(t) \, dt$, and evaluate $\int_0^{3.5} |v(t)| \, dt$. Interpret the meaning of each integral in the context of the problem.
- (d) A second particle moves along the x -axis with position given by $x_2(t) = t^2 - t$ for $0 \leq t \leq 3.5$. At what time t are the two particles moving with the same velocity?

Part A

1 point is earned for answer

$$v'(3) = -2.118$$

The acceleration of the particle at time $t = 3$ is -2.118 .

0	1
---	---

The student earns all of the following points:

1 point is earned for answer

$$v'(3) = -2.118$$

The acceleration of the particle at time $t = 3$ is -2.118 .**Part B**1 point is earned for $\int_0^3 v(t) \, dt$

1 point is earned for using initial condition

1 point is earned for the answer

$$x(3) = x(0) + \int_0^3 v(t) \, dt = -5 + \int_0^3 v(t) \, dt = -1.760213$$

2-9-10

The position of the particle at time $t = 3$ is -1.760 .



0	1	2	3
---	---	---	---

The student earns all of the following point:

1 point is earned for $\int_0^3 v(t) dt$

1 point is earned for using initial condition

1 point is earned for the answer

$$x(3) = x(0) + \int_0^3 v(t) dt = -5 + \int_0^3 v(t) dt = -1.760213$$

The position of the particle at time $t = 3$ is -1.760 .

Part C

1 point is earned for the answers

2 points are earned for the interpretations of $\int_0^{3.5} v(t) dt$ and $\int_0^{3.5} |v(t)| dt$

$$\int_0^{3.5} v(t) dt = 2.844 \text{ (or } 2.843\text{)}$$

$$\int_0^{3.5} |v(t)| dt = 3.737$$

The integral $\int_0^{3.5} v(t) dt$ is the displacement of the particle over the time interval $0 \leq t \leq 3.5$.

The integral $\int_0^{3.5} |v(t)| dt$ is the total distance traveled by the particle over the time interval $0 \leq t \leq 3.5$.

2-9-10



0	1	2	3
---	---	---	---

The student earns all of the following point:

1 point is earned for the answers

2 points are earned for the interpretations of $\int_0^{3.5} v(t) \, dt$ and $\int_0^{3.5} |v(t)| \, dt$

$$\int_0^{3.5} v(t) \, dt = 2.844 \text{ (or 2.843)}$$

$$\int_0^{3.5} |v(t)| \, dt = 3.737$$

The integral $\int_0^{3.5} v(t) \, dt$ is the displacement of the particle over the time interval $0 \leq t \leq 3.5$.

The integral $\int_0^{3.5} |v(t)| \, dt$ is the total distance traveled by the particle over the time interval $0 \leq t \leq 3.5$.

Part D

1 point is earned if sets $v(t) = x_2'(t)$

1 point is earned for the answer

$$v(t) = x_2'(t)$$

$$v(t) = 2t - 1 \Rightarrow t = 1.57054$$

The two particles are moving with the same velocity at time $t = 1.571$ (or 1570).

2-9-10



0	1	2
---	---	---

The student earns all of the following point:

1 point is earned if sets $v(t) = x_2'(t)$

1 point is earned for the answer

$$v(t) = x_2'(t)$$

$$v(t) = 2t - 1 \Rightarrow t = 1.57054$$

The two particles are moving with the same velocity at time $t = 1.571$ (or 1570).

2-9-10

33. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Particle P moves along the y -axis so that its position at time t is given by $y(t) = 4t - \frac{2}{3}$ for all times t . A second particle, particle Q , moves along the x -axis so that its position at time t is given by $x(t) = \frac{\sin(\pi t)}{2-t}$ for all times $t \neq 2$.

(a) As time t approaches 2, what is the limit of the position of particle Q ? Show the work that leads to your answer.

(b) Show that the velocity of particle Q is given by $v_Q(t) = \frac{2\pi \cos(\pi t) - \pi t \cos(\pi t) + \sin(\pi t)}{(2-t)^2}$ for all times $t \neq 2$.

(c) Find the rate of change of the distance between particle P and particle Q at time $t = \frac{1}{2}$. Show the work that leads to your answer.

Part A

The first point requires showing that an indeterminate form $\frac{0}{0}$ is present in order to apply L'Hospital's Rule. Any error in mathematical communication (e.g., writing " $= \frac{0}{0}$ " or not using limit notation) impacts the second point.

The second point does not require a simplified answer. Trigonometric function values do not need to be evaluated. Any differentiation or computation error impacts this point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ application of L'Hospital's Rule
- ☐ answer

Solution:

$$\lim_{t \rightarrow 2} \sin(\pi t) = 0$$

2-9-10

$$\lim_{t \rightarrow 2} (2 - t) = 0$$

By L'Hospital's Rule,

$$\lim_{t \rightarrow 2} \frac{\sin(\pi t)}{2 - t} = \lim_{t \rightarrow 2} \frac{\pi \cos(\pi t)}{-1} = -\pi$$

Part B

The first and second points require evidence of quotient rule and chain rule and no errors. At most 1 out of 2 points is earned for partial communication of quotient rule and chain rule with a maximum of one computational error.

The third point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ quotient rule
- ☐ chain rule
- ☐ verification

Solution:

$$v_Q(t) = x'(t) = \frac{(2-t)\pi \cos(\pi t) - (-1)\sin(\pi t)}{(2-t)^2} = \frac{2\pi \cos(\pi t) - \pi t \cos(\pi t) + \sin(\pi t)}{(2-t)^2}$$

Part C

For the second point, trigonometric function values do not need to be evaluated.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3	4
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The student response accurately includes all four of the criteria below.

- ☐ $y'(\frac{1}{2})$

2-9-10

- ☐ $x' \left(\frac{1}{2} \right)$
- ☐ related rates equation $2D \frac{dD}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$
- ☐ answer

At time $t = \frac{1}{2}$, particle P has position $y \left(\frac{1}{2} \right) = 4 \left(\frac{1}{2} \right) - \frac{2}{3} = 2 - \frac{2}{3} = \frac{4}{3}$ and velocity $y' \left(\frac{1}{2} \right) = v_P \left(\frac{1}{2} \right) = 4$.

At time $t = \frac{1}{2}$, particle Q has position $x \left(\frac{1}{2} \right) = \frac{\sin \left(\frac{\pi}{2} \right)}{2 - \frac{1}{2}} = \frac{1}{\frac{3}{2}} = \frac{2}{3}$ and velocity

$$x' \left(\frac{1}{2} \right) = v_Q \left(\frac{1}{2} \right) = \frac{2\pi \cos \left(\frac{\pi}{2} \right) - \pi \left(\frac{1}{2} \right) \cos \left(\frac{\pi}{2} \right) + \sin \left(\frac{\pi}{2} \right)}{\left(2 - \frac{1}{2} \right)^2} = \frac{1}{\left(\frac{3}{2} \right)^2} = \frac{4}{9}.$$

Let D represent the distance between particle P and particle Q . Using the Pythagorean Theorem, $D^2 = x^2 + y^2$.

Therefore, $2D \frac{dD}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$.

$$\frac{dD}{dt} = \frac{2x \frac{dx}{dt} + 2y \frac{dy}{dt}}{2D} = \frac{x \frac{dx}{dt} + y \frac{dy}{dt}}{D}$$

At time $t = \frac{1}{2}$,

$$D^2 = \left(\frac{2}{3} \right)^2 + \left(\frac{4}{3} \right)^2 \Rightarrow D^2 = \frac{20}{9} \Rightarrow D = \frac{\sqrt{20}}{3}.$$

$$\left. \frac{dD}{dt} \right|_{t=\frac{1}{2}} = \frac{\left(\frac{2}{3} \right) \left(\frac{4}{9} \right) + \left(\frac{4}{3} \right) (4)}{\frac{\sqrt{20}}{3}} = \frac{\frac{8}{27} + \frac{16}{3}}{\frac{\sqrt{20}}{3}} = \frac{152}{27} \cdot \frac{3}{\sqrt{20}} = \frac{152}{9\sqrt{20}}$$

The distance between the particles is increasing at a rate of $\frac{152}{9\sqrt{20}}$.

2-9-10

34. $\frac{d}{dx}(\tan x) =$

(A) $-\cot x$

(B) $-\csc^2 x$

(C) $\cot x$

(D) $\sec^2 x$ ✓

Answer D

Correct. The quotient rule is used as follows.

$$\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{\frac{d}{dx}(\sin x) \cos x - \sin x \frac{d}{dx}(\cos x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$$

35. $\frac{d}{dx}(\csc x) =$

(A) $\sec x$

(B) $-\sec x$

(C) $\csc x \cot x$

(D) $-\csc x \cot x$ ✓

Answer D

Correct. The quotient rule is used as follows.

$$\frac{d}{dx}(\csc x) = \frac{d}{dx}\left(\frac{1}{\sin x}\right) = \frac{0 \cdot \sin x - 1 \cdot (\cos x)}{\sin^2 x} = \frac{-\cos x}{\sin^2 x} = -\frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} = -\csc x \cot x$$

2-9-10

36. Below is an attempt to derive the derivative of $\sec x$ using the product rule, where x is in the domain of $\sec x$. In which step, if any, does an error first appear?

Step 1: $\sec x \cdot \cos x = 1$

Step 2: $\frac{d}{dx}(\sec x \cdot \cos x) = 0$

Step 3: $\frac{d}{dx}(\sec x) \cdot \cos x - \sec x \cdot \sin x = 0$

Step 4: $\frac{d}{dx}(\sec x) = \frac{\sec x \cdot \sin x}{\cos x} = \sec x \cdot \frac{\sin x}{\cos x} = \sec x \cdot \tan x$

- (A) Step 1
(B) Step 2
(C) Step 3

(D) There is no error.



Answer D

Correct. There is no error.

37. If $f(x) = \tan x$, then $\lim_{x \rightarrow \frac{\pi}{4}} \frac{f(x) - f(\frac{\pi}{4})}{x - \frac{\pi}{4}}$ is

- (A) 0
(B) $\tan\left(\frac{\pi}{4}\right)$

(C) $\sec^2\left(\frac{\pi}{4}\right)$

(D) nonexistent



Answer C

Correct. The limit expression is the definition of the derivative of $\tan x$ at $x = \frac{\pi}{4}$.

$$\left. \frac{d}{dx}(\tan x) \right|_{x=\frac{\pi}{4}} = \sec^2 x \Big|_{x=\frac{\pi}{4}}$$

2-9-10

38. $\frac{d}{dx}(\cot x) =$

(A) $-\tan x$

(B) $-\csc^2 x$

(C) $\csc x$

(D) $\sec^2 x$

**Answer B**

Correct. The cotangent function is rewritten as $\frac{\cos x}{\sin x}$, and the quotient rule is used to determine the derivative, as follows.

$$\frac{d}{dx}(\cot x) = \frac{d}{dx}\left(\frac{\cos x}{\sin x}\right) = \frac{\frac{d}{dx}(\cos x) \cdot \sin x - \cos x \cdot \frac{d}{dx}(\sin x)}{(\sin x)^2} = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = \frac{-1}{\sin^2 x} = -\csc^2 x$$

39. Which of the following correctly shows the derivation of $\frac{d}{dx}(\sec x)$?

(A) $\frac{d}{dx}(\sec x) = \frac{d}{dx}\left(\frac{1}{\cos x}\right) = \frac{1}{\frac{d}{dx}(\cos x)} = \frac{1}{\sin x}$

(B) $\frac{d}{dx}(\sec x) = \frac{d}{dx}\left(\frac{1}{\cos x}\right) = \frac{1}{\frac{d}{dx}(\cos x)} = \frac{1}{-\sin x}$

(C) $\frac{d}{dx}(\sec x) = \frac{d}{dx}\left(\frac{1}{\cos x}\right) = \frac{\frac{d}{dx}(1) \cdot \cos x - 1 \cdot \frac{d}{dx}(\cos x)}{(\cos x)^2} = \frac{0 \cdot \cos x - 1 \cdot (-\sin x)}{(\cos x)^2}$

(D) $\frac{d}{dx}(\sec x) = \frac{d}{dx}\left(\frac{1}{\cos x}\right) = \frac{\frac{d}{dx}(1) \cdot \cos x + 1 \cdot \frac{d}{dx}(\cos x)}{(\cos x)^2} = \frac{0 \cdot \cos x + 1 \cdot (-\sin x)}{(\cos x)^2}$

**Answer C**

Correct. The secant function is rewritten as $\frac{1}{\cos x}$, and the quotient rule is used to determine the derivative. Note that

$$\frac{0 \cdot \cos x - 1 \cdot (-\sin x)}{(\cos x)^2} = \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} = \sec x \tan x.$$

2-9-10

40. $\frac{d}{dx}(\cos x \tan x) =$

(A) $\sec x + \sin x \tan x$

(B) $\cos x$ ✓

(C) $-\sin x \sec^2 x$

(D) $\sin x$

Answer B

Correct. It follows from trigonometric identities that $\cos x \tan x = \sin x$. Therefore,

$$\frac{d}{dx}(\cos x \tan x) = \frac{d}{dx}(\sin x) = \cos x.$$

41. Which of the following correctly shows the derivation of $\frac{d}{dx}(\cot x)$?

(A) $\frac{d}{dx}(\cot x) = \frac{d}{dx}\left(\frac{1}{\tan x}\right) = \frac{1}{\frac{d}{dx}(\tan x)} = \frac{-1}{\sec^2 x}$

(B) $\frac{d}{dx}(\cot x) = \frac{d}{dx}\left(\frac{1}{\tan x}\right) = \frac{1}{\frac{d}{dx}(\tan x)} = \frac{1}{\sec^2 x}$

(C) $\frac{d}{dx}(\cot x) = \frac{d}{dx}\left(\frac{1}{\tan x}\right) = \frac{\tan x \frac{d}{dx}(1) - 1 \cdot \frac{d}{dx}(\tan x)}{\tan^2 x} = \frac{(\tan x) \cdot 0 - \sec^2 x}{\tan^2 x} = -\frac{\sec^2 x}{\tan^2 x}$ ✓

(D) $\frac{d}{dx}(\cot x) = \frac{d}{dx}\left(\frac{1}{\tan x}\right) = \frac{\tan x \frac{d}{dx}(1) + 1 \cdot \frac{d}{dx}(\tan x)}{\tan^2 x} = \frac{\tan x \cdot 0 + \sec^2 x}{\tan^2 x} = \frac{\sec^2 x}{\tan^2 x}$

Answer C

Correct. The cotangent function is rewritten as $\frac{1}{\tan x}$, and the quotient rule is used to determine the derivative. Note that

$$-\frac{\sec^2 x}{\tan^2 x} = -\frac{\frac{1}{\cos^2 x}}{\frac{\sin^2 x}{\cos^2 x}} = -\frac{1}{\sin^2 x} = -\csc^2 x.$$

2-9-10

42. If $f(x) = \sec x$, then $\lim_{x \rightarrow \frac{\pi}{3}} \frac{f(x) - f(\frac{\pi}{3})}{x - \frac{\pi}{3}}$ is

(A) 0
(B) $\sec\left(\frac{\pi}{3}\right)$
(C) $\sec\left(\frac{\pi}{3}\right) \tan\left(\frac{\pi}{3}\right)$ ✓
(D) nonexistent

Answer C

Correct. The limit expression is the definition of the derivative of $\sec x$ at $x = \frac{\pi}{3}$.

$$\left. \frac{d}{dx} \sec x \right|_{x=\frac{\pi}{3}} = \sec x \tan x \Big|_{x=\frac{\pi}{3}} = \sec\left(\frac{\pi}{3}\right) \tan\left(\frac{\pi}{3}\right)$$

43. Below is an attempt to derive the derivative of $\csc x$ using the product rule, where x is in the domain of $\csc x$. In which step, if any, does an error first appear?

Step 1: $\csc x \cdot \sin x = 1$

Step 2: $\frac{d}{dx}(\csc x \cdot \sin x) = 0$

Step 3: $\frac{d}{dx}(\csc x) \cdot \sin x + \csc x \cdot \cos x = 0$

Step 4: $\frac{d}{dx}(\csc x) = \frac{-\csc x \cdot \cos x}{\sin x} = -\csc x \cdot \cot x$

(A) Step 1
(B) Step 2
(C) Step 3
(D) There is no error. ✓

Answer D

Correct. There is no error.

2-9-10

44. $\frac{d}{dx}(\sin x \cdot \csc x) =$

(A) 0

(B) 1

(C) $-\cot^2 x$

(D) $2 \cot x$

**Answer A**

Correct. It follows from trigonometric identities that $\sin x \cdot \csc x = 1$. Therefore, $\frac{d}{dx}(\sin x \cdot \csc x) = \frac{d}{dx}(1) = 0$.

45. $\frac{d}{dx}(\cot x) =$

(A) $-\tan x$

(B) $\sec^2 x$

(C) $\tan x$

(D) $-\csc^2 x$

**Answer D**

Correct. The quotient rule is used as follows.

$$\frac{d}{dx}(\cot x) = \frac{d}{dx}\left(\frac{\cos x}{\sin x}\right) = \frac{\frac{d}{dx}(\cos x) \sin x - \frac{d}{dx}(\sin x)(\cos x)}{\sin^2 x} = \frac{-\sin^2 x - \cos^2 x}{\sin^2 x} = \frac{-1}{\sin^2 x} = -\csc^2 x$$

2-9-10

46. $\lim_{x \rightarrow 0} \frac{\sin x}{e^x - 1}$ is

(A) 1

(B) $\frac{1}{e}$

(C) 0

(D) nonexistent

**Answer A**

Correct. Since $\lim_{x \rightarrow 0} \sin x = 0$ and $\lim_{x \rightarrow 0} (e^x - 1) = 0$, the indeterminate limit can be evaluated using L'Hospital's Rule, as follows.

$$\lim_{x \rightarrow 0} \frac{\sin x}{e^x - 1} = \lim_{x \rightarrow 0} \frac{\cos x}{e^x} = \frac{\cos 0}{e^0} = \frac{1}{1} = 1$$

47. If $f(x) = \frac{2x^2 - 1}{5x + 3}$, then $f'(-1) =$

(A) $-\frac{3}{2}$

(B) $-\frac{4}{5}$

(C) $\frac{3}{4}$

(D) $\frac{13}{4}$

**Answer C**

Correct. This question involves the use of the quotient rule to calculate the derivative of a function at a specific point.

$$f'(x) = \frac{4x(5x+3) - 5(2x^2-1)}{(5x+3)^2} = \frac{10x^2 + 12x + 5}{(5x+3)^2}$$
$$f'(-1) = \frac{10 - 12 + 5}{(-2)^2} = \frac{3}{4}$$

2-9-10

48. If $f(x) = \frac{\sin x}{e^x}$, then $f'(x) =$

(A) $\frac{-\cos x - \sin x}{e^x}$

(B) $\frac{\cos x - \sin x}{e^x}$ ✓

(C) $\frac{\sin x - \cos x}{e^x}$

(D) $\frac{\cos x + \sin x}{e^x}$

Answer B

Correct. The quotient rule is applied to determine the derivative, as follows.

$$f'(x) = \frac{\frac{d}{dx}(\sin x) \cdot e^x - \sin x \cdot \frac{d}{dx}(e^x)}{(e^x)^2} = \frac{\cos x \cdot e^x - \sin x \cdot e^x}{(e^x)^2} = \frac{\cos x - \sin x}{e^x}$$

49. If $f(x) = \frac{3x^2 - 2}{4x + 1}$, then $f'(-1) =$

(A) $-\frac{14}{3}$

(B) $-\frac{3}{2}$

(C) $\frac{14}{9}$ ✓

(D) $\frac{22}{9}$

Answer C

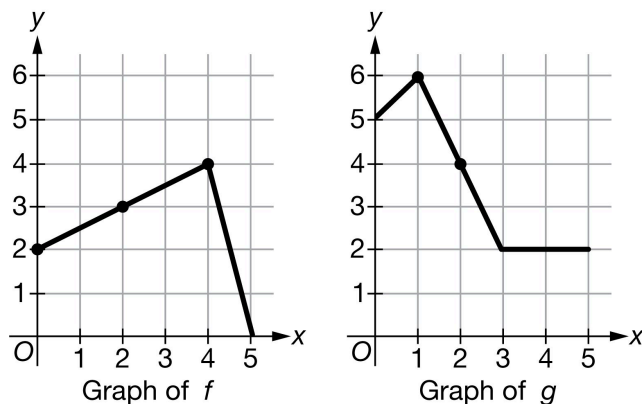
Correct. This question involves the use of the quotient rule to calculate the derivative of a function at a specific point.

$$f'(x) = \frac{(4x+1)(6x) - (3x^2-2)(4)}{(4x+1)^2} = \frac{12x^2+6x+8}{(4x+1)^2}$$

$$f'(-1) = \frac{12-6+8}{(-4+1)^2} = \frac{14}{9}$$

2-9-10

50.



The graphs of the functions f and g are shown above. If $h(x) = \frac{f(x)+1}{g(x)+3x}$, then $h'(2) =$

- (A) $\frac{1}{2}$
 (B) $\frac{9}{100}$
 (C) $\frac{1}{100}$ ✓
 (D) $\frac{1}{10}$

Answer C

Correct. Use the quotient rule to calculate $h'(x)$, then evaluate at $x = 2$.

$$h'(x) = \frac{(g(x)+3x)f'(x) - (f(x)+1)(g'(x)+3)}{(g(x)+3x)^2}$$

$$h'(2) = \frac{(g(2)+3(2))f'(2) - (f(2)+1)(g'(2)+3)}{(g(2)+3(2))^2}$$

The graph of f is used to determine that $f(2) = 3$ and $f'(2) = \frac{1}{2}$. The graph of g is used to determine that $g(2) = 4$ and $g'(2) = -2$.

$$\text{Therefore, } h'(2) = \frac{(4+6)(\frac{1}{2}) - (3+1)(-2+3)}{(4+6)^2} = \frac{5-4}{10^2} = \frac{1}{100}.$$

2-9-10

51. What is the slope of the line tangent to the graph of $y = \frac{4x^3}{x+3}$ at $x = 1$?

(A) 1

(B) $\frac{11}{4}$

(C) $\frac{13}{4}$

(D) 12



Answer B

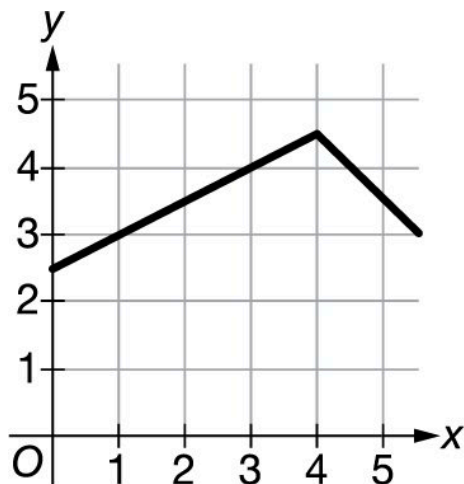
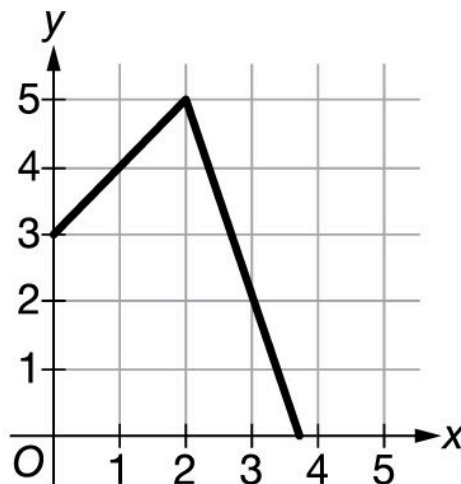
Correct. The slope of the line tangent to the graph at $x = 1$ is the value of the derivative of $\frac{4x^3}{x+3}$ with respect to x at $x = 1$.

The quotient rule can be used to differentiate $\frac{4x^3}{x+3}$, as follows.

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \left(\frac{4x^3}{x+3} \right) = \frac{\frac{d}{dx}(4x^3)(x+3) - 4x^3 \frac{d}{dx}(x+3)}{(x+3)^2} = \frac{12x^2(x+3) - 4x^3(1)}{(x+3)^2} = \frac{8x^3 + 36x^2}{(x+3)^2} \\ \frac{dy}{dx} \Big|_{x=1} &= \frac{8x^3 + 36x^2}{(x+3)^2} \Big|_{x=1} = \frac{8 \cdot 1^3 + 36 \cdot 1^2}{(1+3)^2} = \frac{8+36}{4^2} = \frac{44}{16} = \frac{11}{4}\end{aligned}$$

2-9-10

52.

Graph of f Graph of g

The graphs of the functions f and g are shown above. If $h(x) = \frac{f(x)+4}{g(x)+2x}$, then $h'(3) =$

- (A) $-\frac{1}{2}$
 (B) $-\frac{1}{16}$
 (C) $\frac{3}{16}$
 (D) $\frac{3}{2}$



Answer C

Correct. Use the quotient rule to calculate $h'(x)$, then evaluate at $x = 3$.

$$h'(x) = \frac{f'(x)(g(x)+2x) - (f(x)+4)(g'(x)+2)}{(g(x)+2x)^2}$$

$$h'(3) = \frac{f'(3)(g(3)+6) - (f(3)+4)(g'(3)+2)}{(g(3)+6)^2}$$

The graph of f is used to determine that $f(3) = 4$ and $f'(3) = \frac{1}{2}$. The graph of g is used to determine that $g(3) = 2$ and $g'(3) = -3$. Therefore, $h'(3) = \frac{\frac{1}{2}(2+6) - (4+4)(-3+2)}{(2+6)^2} = \frac{4+8}{64} = \frac{3}{16}$.

2-9-10

53. Let f be a differentiable function such that $f(9) = 18$ and $f'(9) = 7$. If g is the function defined by $g(x) = \frac{f(x)}{\sqrt{x}}$, what is the value of $g'(9)$?

(A) 2

(B) $\frac{7}{3}$ (C) $\frac{8}{3}$

(D) 42

Answer A

Correct. The quotient rule is applied to determine the derivative, as follows.

$$g'(x) = \frac{f'(x)\sqrt{x} - f(x)\frac{d}{dx}(\sqrt{x})}{(\sqrt{x})^2} = \frac{f'(x)\sqrt{x} - f(x)\frac{1}{2\sqrt{x}}}{x}$$

$$g'(9) = \frac{f'(9)\sqrt{9} - f(9)\frac{1}{2\sqrt{9}}}{9} = \frac{7 \cdot 3 - 18 \cdot \frac{1}{6}}{9} = 2$$

54. If $f(x) = \frac{\cos x}{\ln x}$, then $f'(x) =$

(A) $\frac{x \sin x \ln x - \cos x}{x(\ln x)^2}$ (B) $\frac{-x \sin x \ln x - \cos x}{x(\ln x)^2}$ (C) $-x \sin x$ (D) $\frac{-x \sin x \ln x + \cos x}{x(\ln x)^2}$ **Answer B**

Correct. The quotient rule is applied to determine the derivative, as follows.

$$f'(x) = \frac{\frac{d}{dx}(\cos x) \cdot \ln x - \cos x \cdot \frac{d}{dx}(\ln x)}{(\ln x)^2} = \frac{-\sin x \ln x - \cos x \left(\frac{1}{x}\right)}{(\ln x)^2} = \frac{-x \sin x \ln x - \cos x}{x(\ln x)^2}$$

2-9-10

55. Let f be a differentiable function such that $f(8) = 2$ and $f'(8) = 5$. If g is the function defined by $g(x) = \frac{f(x)}{\sqrt[3]{x}}$, what is the value of $g'(8)$?

(A) $\frac{59}{24}$

(B) $\frac{5}{2}$

(C) $\frac{61}{24}$

(D) 60

**Answer A**

Correct. The quotient rule is applied to determine the derivative, as follows.

$$g'(x) = \frac{f'(x) \cdot x^{\frac{1}{3}} - f(x) \cdot \frac{d}{dx} \left(x^{\frac{1}{3}} \right)}{\left(x^{\frac{1}{3}} \right)^2} = \frac{f'(x) x^{\frac{1}{3}} - f(x) \left(\frac{1}{3} x^{-\frac{2}{3}} \right)}{x^{\frac{2}{3}}}$$

$$g'(8) = \frac{f'(8) \left(8^{\frac{1}{3}} \right) - f(8) \left(\frac{1}{3} \left(8^{-\frac{2}{3}} \right) \right)}{8^{\frac{2}{3}}} = \frac{5 \cdot 2 - 2 \cdot \left(\frac{1}{3 \cdot 4} \right)}{4} = \frac{5 \cdot 2 \cdot 3 \cdot 4 - 2}{3 \cdot 4 \cdot 4} = \frac{120 - 2}{48} = \frac{118}{48} = \frac{59}{24}$$

56.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
0	4	$\frac{1}{2}$	-2	$\frac{3}{2}$

The table above gives values of the differentiable functions f and g and their derivatives at $x = 0$. If

$$h(x) = \frac{6f(x)}{g(x)-1}, \text{ then } h'(0) =$$

(A) 15

(B) 3

(C) 2

(D) -5

**Answer D**

Correct. The quotient rule is applied to determine the derivative, as follows.

$$h'(x) = \frac{\frac{d}{dx}(6f(x))(g(x)-1) - 6f(x) \frac{d}{dx}(g(x)-1)}{(g(x)-1)^2} = \frac{6f'(x)(g(x)-1) - 6f(x)g'(x)}{(g(x)-1)^2}$$

$$h'(0) = \frac{6f'(0)(g(0)-1) - 6f(0)g'(0)}{(g(0)-1)^2} = \frac{6 \cdot \frac{1}{2} \cdot (-2-1) - 6 \cdot (4) \cdot \frac{3}{2}}{(-2-1)^2} = \frac{-9-36}{9} = \frac{-45}{9} = -5$$

2-9-10

57. If $f(x) = x^2 + 4x - 2$, $g(x) = 4x + 2$, and $h(x) = \frac{f(x)}{g(x)}$, then $h'(1) =$

(A) $-\frac{2}{3}$

(B) $\frac{2}{3}$

(C) $\frac{4}{3}$

(D) $\frac{3}{2}$

Answer B

Correct. The derivative of h is found using the quotient rule.

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

$$f(1) = 3; f'(x) = 2x + 4 \Rightarrow f'(1) = 6$$

$$g(1) = 6; g'(x) = 4 \Rightarrow g'(1) = 4$$

Therefore,

$$h'(1) = \frac{(6)(6) - (3)(4)}{6^2} = \frac{36 - 12}{36} = \frac{24}{36} = \frac{2}{3}.$$

58.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	5	3	2	4

The table shown gives values of the differentiable functions f and g and their derivatives at $x = 2$. If

$h(x) = \frac{f(x)}{g(x)}$, what is the value of $h'(2)$?

(A) $-\frac{7}{2}$

(B) $-\frac{7}{8}$

(C) $\frac{3}{4}$

(D) $\frac{13}{2}$

Answer A

Correct. The derivative of h is found using the quotient rule.

$$h'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

$$h'(2) = \frac{f'(2)g(2) - f(2)g'(2)}{(g(2))^2} = \frac{3(2) - 5(4)}{2^2} = \frac{6 - 20}{4} = -\frac{14}{4} = -\frac{7}{2}$$

2-9-10

59. If $y = \frac{x^3}{x+2}$, then $\frac{dy}{dx} =$

(A) $3x^2$

(B) $\frac{-2x^3-6x^2}{(x+2)^2}$

(C) $\frac{2x^3+6x^2}{(x+2)^2}$

(D) $\frac{4x^3+6x^2}{(x+2)^2}$

**Answer C**

Correct. The derivative is found using the quotient rule.

$$\begin{aligned}\frac{dy}{dx} &= \frac{\frac{d}{dx}(x^3) \cdot (x+2) - (x^3) \cdot \frac{d}{dx}(x+2)}{(x+2)^2} \\ &= \frac{(3x^2)(x+2) - (x^3)(1)}{(x+2)^2} = \frac{3x^3+6x^2-x^3}{(x+2)^2} = \frac{2x^3+6x^2}{(x+2)^2}\end{aligned}$$

60.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
2	-1	-4	5	3

The table shown gives values of functions f , g , f' , and g' at $x = 2$. If $h(x) = \frac{f(x)}{g(x)}$, then $h'(2) =$

(A) $-\frac{23}{16}$

(B) $-\frac{17}{16}$

(C) $-\frac{11}{16}$

(D) $\frac{5}{3}$

**Answer B**Correct. The derivative of h is found using the quotient rule.

$$\begin{aligned}h'(x) &= \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2} \\ h'(2) &= \frac{g(2)f'(2) - f(2)g'(2)}{(g(2))^2} = \frac{(-4)(5) - (-1)(3)}{(-4)^2} = \frac{-20+3}{16} = -\frac{17}{16}\end{aligned}$$

2-9-10

61.

x	$f(x)$	$f'(x)$	$f''(x)$	$g(x)$	$g'(x)$	$g''(x)$
2	4	-3	3	-2	5	1

The table above gives values of the twice-differentiable functions f and g and their derivatives at $x = 2$.

If h is the function defined by $h(x) = \frac{f'(x)}{g(x)}$, what is the value of $h'(2)$?

(A) $\frac{9}{4}$

(B) $\frac{3}{5}$

(C) $-\frac{3}{2}$

(D) $-\frac{21}{4}$

**Answer A**

Correct. The derivative of h is found by using the quotient rule.

$$h'(x) = \frac{f''(x)g(x) - f'(x)g'(x)}{(g(x))^2}$$

The values for the functions and derivatives at $x = 2$ are obtained from the table.

$$h'(2) = \frac{f''(2)g(2) - f'(2)g'(2)}{(g(2))^2} = \frac{(3)(-2) - (-3)(5)}{(-2)^2} = \frac{-6+15}{4} = \frac{9}{4}$$

2-9-10

62.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	-3	$\frac{1}{2}$	-3	2

The table above gives values of the differentiable functions f and g and their derivatives at $x = 2$. If

$$h(x) = \frac{4f(x)}{g(x)+1}, \text{ then } h'(2) =$$

- (A) -10
 (B) -7
 (C) 1
 (D) 5



Answer D

Correct. The quotient rule is applied to determine the derivative, as follows.

$$h'(x) = \frac{\frac{d}{dx}(4f(x))(g(x)+1) - 4f(x)\frac{d}{dx}(g(x)+1)}{(g(x)+1)^2} = \frac{4f'(x)(g(x)+1) - 4f(x)g'(x)}{(g(x)+1)^2}$$

$$h'(2) = \frac{4f'(2)(g(2)+1) - 4f(2)g'(2)}{(g(2)+1)^2} = \frac{4 \cdot \frac{1}{2} \cdot (-3+1) - 4 \cdot (-3) \cdot 2}{(-3+1)^2} = \frac{-4+24}{4} = 5$$

63. What is the slope of the line tangent to the graph of $y = \frac{9x^2}{x+2}$ at $x = 1$?

- (A) 3
 (B) 5
 (C) 7
 (D) 18



Answer B

Correct. The slope of the line tangent to the graph at $x = 1$ is the value of the derivative of $\frac{9x^2}{x+2}$ with respect to x at $x = 1$. The quotient rule can be used to differentiate $\frac{9x^2}{x+2}$, as follows.

$$\frac{dy}{dx} = \frac{\frac{d}{dx}(9x^2)(x+2) - 9x^2\frac{d}{dx}(x+2)}{(x+2)^2} = \frac{18x(x+2) - 9x^2 \cdot 1}{(x+2)^2} = \frac{9x^2 + 36x}{(x+2)^2}$$

$$\left. \frac{dy}{dx} \right|_{x=1} = \frac{9x^2 + 36x}{(x+2)^2} \Big|_{x=1} = \frac{9 \cdot 1^2 + 36 \cdot 1}{(1+2)^2} = \frac{45}{9} = 5$$

2-9-10

64. The function f is given by $f(x) = \frac{\sin x}{x}$. What is the instantaneous rate of change of f at $x = \frac{\pi}{2}$?

(A) $-\frac{4}{\pi^2}$



(B) 0

(C) $\frac{2}{\pi}$

(D) $\frac{4}{\pi^2}$

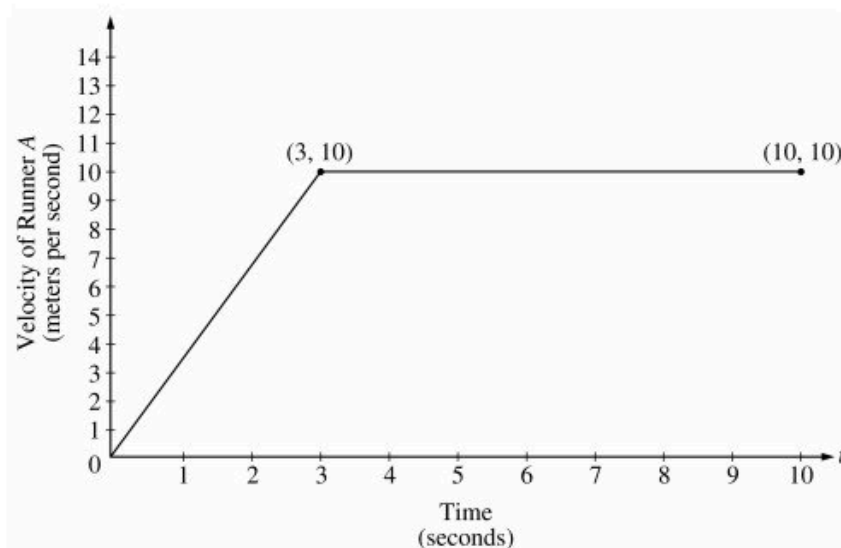
Answer A

Correct. The instantaneous rate of change of f at $x = \frac{\pi}{2}$ is $f'\left(\frac{\pi}{2}\right)$. The quotient rule is used to find the derivative.

$$f'(x) = \frac{x \cos x - \sin x}{x^2} \Rightarrow f'\left(\frac{\pi}{2}\right) = \frac{\frac{\pi}{2} \cdot 0 - 1}{\left(\frac{\pi}{2}\right)^2} = -\frac{4}{\pi^2}$$

2-9-10

65.



Two runners, A and B , run on a straight racetrack for $0 \leq t \leq 10$ seconds. The graph above, which consists of two line segments, shows the velocity, in meters per second, of Runner A . The velocity, in meters per second, of Runner B is given by the function v defined by $v(t) = \frac{24t}{2t+3}$.

- Find the velocity of Runner A and the velocity of Runner B at time $t = 2$ seconds. Indicate units of measure.
- Find the acceleration of Runner A and the acceleration of Runner B at time $t = 2$ seconds. Indicate units of measure.
- Find the total distance run by Runner A and the total distance run by Runner B over the time interval $0 \leq t \leq 10$ seconds. Indicate units of measure.

Part A

1 point is earned for the velocity for Runner A

1 point is earned for the velocity for Runner B

$$\begin{aligned} \text{Runner } A: \text{ velocity} &= \frac{10}{3} \cdot 2 = \frac{20}{3} \\ &= 6.666 \text{ or } 6.667 \text{ meters/sec} \end{aligned}$$

$$\text{Runner } B: v(2) = \frac{48}{7} = 6.857 \text{ meters/sec}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the velocity for Runner A

2-9-10

1 point is earned for the velocity for Runner B

$$\begin{aligned}\text{Runner } A: \text{ velocity} &= \frac{10}{3} \cdot 2 = \frac{20}{3} \\ &= 6.666 \text{ or } 6.667 \text{ meters/sec}\end{aligned}$$

$$\text{Runner } B: v(2) = \frac{48}{7} = 6.857 \text{ meters/sec}$$

Part B

1 point is earned for the acceleration for Runner A

1 point is earned for the acceleration for Runner B

$$\text{Runner } A: \text{ acceleration} = \frac{10}{3} = 3.333 \text{ meters/sec}^2$$

$$\begin{aligned}\text{Runner } B: a(2) = v'(2) &= \frac{72}{(2t+3)^2} \Big|_{t=2} \\ &= \frac{72}{49} = 1.469 \text{ meters/sec}^2\end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the acceleration for Runner A

1 point is earned for the acceleration for Runner B

$$\text{Runner } A: \text{ acceleration} = \frac{10}{3} = 3.333 \text{ meters/sec}^2$$

$$\begin{aligned}\text{Runner } B: a(2) = v'(2) &= \frac{72}{(2t+3)^2} \Big|_{t=2} \\ &= \frac{72}{49} = 1.469 \text{ meters/sec}^2\end{aligned}$$

Part C

2 points are earned for distance for Runner A

1 : method

1 : answer

2-9-10

2 points are earned for distance for Runner *B*

1 : integral

1 : answer

Runner *A*: distance = $\frac{1}{2}(3)(10) + 7(10) = 85$ meters

Runner *B*: distance = $\int_0^{10} \frac{24t}{2t+3} dt = 83.336$ meters

1 point is earned for the units

(units) meters/sec in part (a), meters/sec² in part (b), and meters in part (c), or equivalent.



0	1	2	3	4	5
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The student response earns all of the following points:

2 points are earned for distance for Runner *A*

1 : method

1 : answer

2 points are earned for distance for Runner *B*

1 : integral

1 : answer

Runner *A*: distance = $\frac{1}{2}(3)(10) + 7(10) = 85$ meters

Runner *B*: distance = $\int_0^{10} \frac{24t}{2t+3} dt = 83.336$ meters

1 point is earned for the units

(units) meters/sec in part (a), meters/sec² in part (b), and meters in part (c), or equivalent.

2-9-10

66. Let f be the function given by $f(x) = e^{\frac{x}{2}}$.

(a) Write the first four nonzero terms and the general term for the Taylor series expansion of $f(x)$ about $x = 0$.

(b) Use the results from part (a) to write the first three nonzero terms and the general term of the series expansion about $x = 0$ for $g(x) = \frac{e^{\frac{x}{2}} - 1}{x}$.

(c) For the function g in part (b), find $g'(2)$ and use it to show that $\sum_{n=1}^{\infty} \frac{n}{4(n+1)!} = \frac{1}{4}$.

Part A

1 point is earned for first two terms

1 point is earned for second two terms

1 point is earned for general term

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots$$

$$e^{x/2} = 1 + \frac{x}{2} + \frac{(x/2)^2}{2!} + \frac{(x/2)^3}{3!} + \cdots + \frac{(x/2)^n}{n!} + \cdots$$

$$= 1 + \frac{x}{2} + \frac{x^2}{2^2 2!} + \frac{x^3}{2^3 3!} + \cdots + \frac{x^n}{2^n n!} + \cdots$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for first two terms

1 point is earned for second two terms

1 point is earned for general term

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots$$

$$e^{x/2} = 1 + \frac{x}{2} + \frac{(x/2)^2}{2!} + \frac{(x/2)^3}{3!} + \cdots + \frac{(x/2)^n}{n!} + \cdots$$

$$= 1 + \frac{x}{2} + \frac{x^2}{2^2 2!} + \frac{x^3}{2^3 3!} + \cdots + \frac{x^n}{2^n n!} + \cdots$$

Part B

2-9-10

1 point is earned for first three terms

1 point is earned for general term

$$\begin{aligned}\frac{e^{x/2}-1}{x} &= \frac{\frac{x}{2} + \frac{x^2}{2^2 2!} + \frac{x^3}{2^3 3!} + \cdots + \frac{x^n}{2^n n!} + \cdots}{x} \\ &= \frac{1}{2} + \frac{x}{2^2 2!} + \frac{x^2}{2^3 3!} + \cdots + \frac{x^{n-1}}{2^n n!} + \cdots\end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for first three terms

1 point is earned for general term

$$\begin{aligned}\frac{e^{x/2}-1}{x} &= \frac{\frac{x}{2} + \frac{x^2}{2^2 2!} + \frac{x^3}{2^3 3!} + \cdots + \frac{x^n}{2^n n!} + \cdots}{x} \\ &= \frac{1}{2} + \frac{x}{2^2 2!} + \frac{x^2}{2^3 3!} + \cdots + \frac{x^{n-1}}{2^n n!} + \cdots\end{aligned}$$

Part C

1 point is earned for setting equation of $g(x)$

2 points are earned for differential

1 point is earned for $g'(2) = 1/4$

2-9-10

$$g'(x) = \frac{1}{2^2 2!} + \frac{2x}{2^3 3!} + \cdots + \frac{(n-1)x^{n-2}}{2^n n!} + \cdots$$

$$= \frac{1}{8} + \frac{x}{24} + \cdots + \frac{(n-1)x^{n-2}}{2^n n!} + \cdots$$

$$g'(2) = \frac{1}{2^2 2!} + \frac{2 \cdot 2}{2^3 3!} + \cdots + \frac{(n-1)2^{n-2}}{2^n n!} + \cdots$$

$$= \frac{1}{8} + \frac{1}{12} + \cdots + \frac{n-1}{4n!} + \cdots$$

$$= \sum_{n=1}^{\infty} \frac{n}{4(n+1)!}$$

$$\text{Also, } g(x) = \frac{e^{x/2} - 1}{x}$$

$$g'(x) = \frac{x \cdot \frac{1}{2} e^{x/2} - (1)(e^{x/2} - 1)}{x^2}$$

$$g'(2) = \frac{2 \cdot \frac{1}{2} e - (e - 1)}{4} = \frac{1}{4}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for setting equation of $g(x)$

2 points are earned for differential

1 point is earned for $g'(2) = 1/4$

2-9-10

$$g'(x) = \frac{1}{2^2 2!} + \frac{2x}{2^3 3!} + \cdots + \frac{(n-1)x^{n-2}}{2^n n!} + \cdots$$

$$= \frac{1}{8} + \frac{x}{24} + \cdots + \frac{(n-1)x^{n-2}}{2^n n!} + \cdots$$

$$g'(2) = \frac{1}{2^2 2!} + \frac{2 \cdot 2}{2^3 3!} + \cdots + \frac{(n-1)2^{n-2}}{2^n n!} + \cdots$$

$$= \frac{1}{8} + \frac{1}{12} + \cdots + \frac{n-1}{4n!} + \cdots$$

$$= \sum_{n=1}^{\infty} \frac{n}{4(n+1)!}$$

Also, $g(x) = \frac{e^{x/2} - 1}{x}$

$$g'(x) = \frac{x \cdot \frac{1}{2} e^{x/2} - (1)(e^{x/2} - 1)}{x^2}$$

$$g'(2) = \frac{2 \cdot \frac{1}{2} e - (e - 1)}{4} = \frac{1}{4}$$

2-9-10

67. Let f be the function given by $f(x) = \frac{|x|-2}{x-2}$.

(a) Find all the zeros of f .

(b) Find $f'(1)$.

(c) Find $f'(-1)$.

(d) Find the range of f .

Part A

1 point is earned for $x = -2$

$$f(x) = 0 \Leftrightarrow |x| = 2, x \neq 2$$

$$x = -2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for $x = -2$

$$f(x) = 0 \Leftrightarrow |x| = 2, x \neq 2$$

$$x = -2$$

Part B

1 point is earned for $f'(1) = 0$

For $x \geq 0, x \neq 2$

$$f(x) = \frac{x-2}{x-2} = 1$$

Therefore $f'(1) = 0$



0	1
---	---

The student response earns all of the following points:

2-9-10

1 point is earned for $f'(1) = 0$

For $x \geq 0, x \neq 2$

$$f(x) = \frac{x-2}{x-2} = 1$$

Therefore $f'(1) = 0$

Part C

4 points are earned if solves for $f'(x)$ and $f'(-1)$

$$\text{For } x < 0, f(x) = \frac{-x-2}{x-2} = \frac{-(x+2)}{x-2}$$

$$f'(x) = \frac{(x-2)(-1) - (-x-2)(1)}{(x-2)^2} = \frac{4}{(x-2)^2}$$

$$\therefore f'(-1) = \frac{4}{9}$$



0	1	2	3	4
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The student response earns all of the following points:

4 points are earned if solves for $f'(x)$ and $f'(-1)$

$$\text{For } x < 0, f(x) = \frac{-x-2}{x-2} = \frac{-(x+2)}{x-2}$$

$$f'(x) = \frac{(x-2)(-1) - (-x-2)(1)}{(x-2)^2} = \frac{4}{(x-2)^2}$$

$$\therefore f'(-1) = \frac{4}{9}$$

Part D

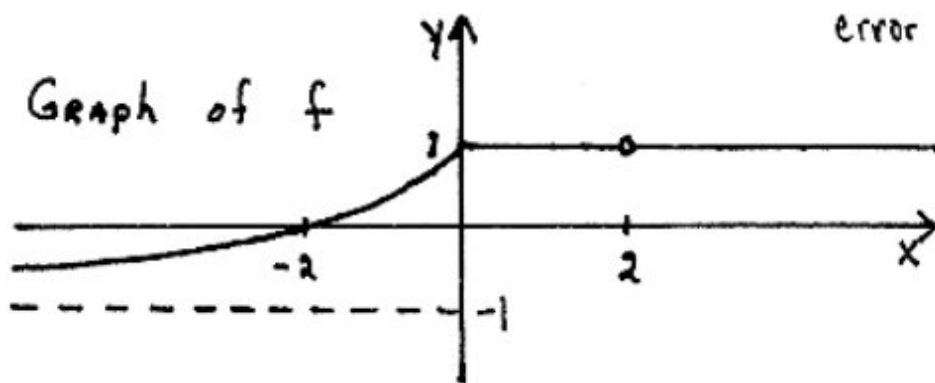
1 point is earned for least upper bound (lub) $y = 1$

1 point is earned for greatest lower bound (glb) $y = -1$

1 point is earned for correct inequalities at -1 and 1

$$-1 < y \leq 1$$

2-9-10



✓

0	1	2	3
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The student response earns all of the following points:

1 point is earned for least upper bound (lub) $y = 1$

1 point is earned for greatest lower bound (glb) $y = -1$

1 point is earned for correct inequalities at -1 and 1

$-1 < y \leq 1$

